Research Square

Preprints are preliminary reports that have not undergone peer review.
They should not be considered conclusive, used to inform clinical practice,
or referenced by the media as validated information.

Cross-domain transfer of frozen 3DINO representations for marker-sparing classification of hepatic 3D nuclear morphotypes

Alisson Terán Ballagán
Facultad de Ingeniería y Ciencias Aplicadas, Universidad de Las Américas
Jean Taylor
Departmento de Biología Celular, Facultad de Ciencias Biológicas, Universidad de Concepción
Dilan Martínez-Torres
Departmento de Biología Celular, Facultad de Ciencias Biológicas, Universidad de Concepción
Fabián Segovia-Miranda
Departmento de Biología Celular, Facultad de Ciencias Biológicas, Universidad de Concepción
Hernán Morales-Navarrete
hernan.morales@udla.edu.ec

Bio-Cheminformatics Research Group and Facultad de Ingeniería y Ciencias Aplicadas, Universidad de Las Américas

Article

Keywords: 3D microscopy, liver tissue, nuclear morphology, DAPI, self-supervised learning, 3DINO, bioimage analysis, machine learning, cell classification, tissue architecture

Posted Date: October 5th, 2026

DOI: https://doi.org/10.21203/rs.3.rs-11238308/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License.
Read Full License

Additional Declarations: No competing interests reported.

Cross-domain transfer of frozen 3DINO representations for marker-sparing classification of hepatic 3D nuclear morphotypes

Alisson Terán Ballagán[1], Jean Taylor[2,3], Dilan Martínez-Torres[2,3], Fabián Segovia-Miranda[2,3,#], Hernán Morales-Navarrete[4,#]

1 Facultad de Ingeniería y Ciencias Aplicadas, Universidad de Las Américas, Quito, 170504, Ecuador
2 Departmento de Biología Celular, Facultad de Ciencias Biológicas, Universidad de Concepción, Concepción, Chile
3 Grupo de Procesos en Biología del Desarrollo (GDeP), Facultad de Ciencias Biológicas, Universidad de Concepción, Concepción, Chile

4 Bio-Cheminformatics Research Group and Facultad de Ingeniería y Ciencias Aplicadas, Universidad de Las Américas, Quito, 170504, Ecuador

Correspondence: fabiansegovia@udec.cl; hernan.morales@udla.edu.ec

Abstract

Cell-type annotation in three-dimensional fluorescence microscopy often relies on dedicated molecular markers. Yet fluorescence channels are limited and often needed to label other structures of interest, so assigning one to a cell-type marker comes at a cost. A nuclear stain such as DAPI, by contrast, is present in nearly all protocols; classifying cell types from DAPI alone would therefore apply broadly to existing datasets while freeing channels for other structures. Here, we tested whether frozen self-supervised volumetric representations can support marker-sparing classification of hepatic 3D nuclear morphotypes from DAPI alone. We analyzed 14,385 nuclei from five adult mouse liver volumes and assigned marker-assisted reference morphotypes comprising hepatocyte, stellate, Kupffer, endothelial, and other cells. Using leave-one-animal-out cross-validation, we compared frozen 3DINO embeddings followed by validation-selected classical classifiers with 31 handcrafted DAPI-derived nuclear features and direct ResNet3D-18 classification. The 3DINO-based procedure achieved the highest mean held-out performance, with macro-F1 of 0.596 ± 0.041, balanced accuracy of 0.625 ± 0.037, and weighted F1 of 0.831 ± 0.046. ResNet3D-18 performed similarly (macro-F1 0.584 ± 0.051), whereas handcrafted features yielded lower mean performance (0.511 ± 0.040). Class-specific analysis showed near-perfect hepatocyte classification but greater ambiguity among non-parenchymal morphotypes. Thus, a frozen volumetric representation pretrained outside fluorescence microscopy achieved similar observed held-out performance to a task-specific 3D CNN trained end-to-end, while requiring only lightweight downstream classifier training. This provides a marker-sparing strategy for hepatic nuclear morphotyping while preserving fluorescence channels for other biological readouts.

Keywords: 3D microscopy; liver tissue; nuclear morphology; DAPI; self-supervised learning; 3DINO; bioimage analysis; machine learning; cell classification; tissue architecture.

Introduction

Quantitative reconstruction of tissue architecture requires reliable identification of the cellular components that form the tissue. In the liver, parenchymal hepatocytes and non-parenchymal cells, including liver sinusoidal endothelial cells, Kupffer cells, and stellate cells, form spatially organized microenvironments that support metabolism, immune surveillance, vascular exchange, and tissue remodeling [1,2]. Robust cell annotation is

therefore a prerequisite for computational analyses of liver organization in development, homeostasis, regeneration, and disease [3,4].

Three-dimensional fluorescence microscopy provides the spatial resolution needed to study liver tissue architecture at cellular and subcellular scales [5–7]. However, cell-type identification commonly depends on fluorescent molecular markers. In practice, the number of fluorescent labels that can be simultaneously distinguished is limited; basic confocal systems typically discriminate only a small number of fluorophores, while conventional spectral fluorescence microscopy is generally restricted to approximately four or five labels [8,9]. This constraint is particularly relevant for volumetric liver imaging, where simultaneous acquisition of multiple fluorescence channels in thick tissue remains experimentally challenging [3]. Consequently, each channel dedicated to cell-type identification reduces the number of channels available for additional structural, functional, signaling, metabolic, extracellular-matrix, or disease-associated readouts.

A marker-sparing alternative is to use cell-type-specific markers during dataset construction while excluding them from the predictive input at inference. In this setting, fluorescence markers, together with morphological and spatial criteria where necessary, provide reference annotations for model development and evaluation, while the deployed classifier predicts marker-associated nuclear morphotypes from a broadly available structural channel. DAPI is particularly attractive for this purpose because it labels nuclear DNA, and nuclear shape, volume, intensity distribution, and chromatin organization can vary across cell types and phenotypic states [10–12]. In liver tissue, quantitative image-derived nuclear and spatial features have previously been used for the automatic recognition and characterization of non-parenchymal cell populations [13]. This approach reconstructs nuclei in 3D and combines morphological and textural descriptors derived from the DNA signal with local information from additional structural markers, such as the sinusoidal endothelial marker FLK1 and the cell-border marker Phalloidin. These measurements can support supervised discrimination of hepatocytes from non-parenchymal cells and further separation of sinusoidal endothelial, Kupffer, and stellate cells [5,13]. However, reliance on additional structural channels limits applicability to datasets in which those signals have been acquired. Classification from DAPI alone is more restrictive because the model must infer marker-associated morphotypes from nuclear information without access to the molecular or neighboring structural signals used to establish the reference annotations.

Handcrafted nuclear descriptors provide interpretable measurements of properties such as size, shape, intensity, and texture, but predefined feature sets may not capture complex three-dimensional combinations of nuclear morphology and chromatin patterning. Self-supervised representation learning offers a complementary strategy by using pretrained models to extract high-dimensional representations without requiring task-specific feature engineering. NuSPIRe recently demonstrated that large-scale self-supervised pretraining on DAPI-stained nuclear images can generate representations useful for cell-type identification and perturbation detection [14]. These findings support the potential of nuclear morphology as a source of discriminative phenotypic information, while leaving open a complementary question: can a volumetric representation pretrained outside the fluorescence-microscopy domain transfer effectively to three-dimensional nuclear phenotyping? 3DINO is a self-supervised framework for volumetric representation learning that adapts DINOv2 to three-dimensional medical imaging and was pretrained on approximately 100,000 volumetric scans spanning more than ten organs [15]. Although developed for medical imaging volumes rather than fluorescence microscopy of individual nuclei, its ability to encode three-dimensional structure makes it a suitable model for testing such cross-domain transfer.

Here, we evaluate whether frozen 3DINO representations can support marker-sparing classification of hepatic 3D nuclear morphotypes from DAPI alone. We developed a leakage-controlled pipeline that segments nuclei using NucVerse3D [16], extracts standardized single-nucleus 3D DAPI patches, assigns marker-assisted reference morphotypes, and compares three strategies under leave-one-animal-out cross-validation: frozen 3DINO embeddings followed by classical machine-learning classifiers, interpretable handcrafted DAPI-derived nuclear features followed by the same classifier families, and direct supervised classification of the DAPI patches using a 3D convolutional neural network (ResNet3D-18). The central objective is to determine whether a frozen

self-supervised volumetric representation pretrained in a different imaging domain can transfer effectively to hepatic nuclear morphotype classification while requiring only the DAPI channel at inference.

Results

Study design and DAPI-only classification framework

We assembled a DAPI-only classification dataset from five high-resolution three-dimensional multiphoton fluorescence microscopy volumes of adult mouse liver tissue, with each volume acquired from a different animal and spanning a central vein–portal vein region of the liver lobule. This central vein–portal vein axis is the repeating structural and functional unit of the liver, along which hepatocytes and non-parenchymal cells show characteristic zonated distributions; sampling it therefore captures the cellular heterogeneity of the hepatic lobule [17,18]. Nuclei were segmented from the DAPI volumes using NucVerse3D [16] and subsequently assigned to five marker-assisted reference morphotypes: hepatocytes, stellate cells, Kupffer cells, endothelial cells, and other cells (Fig. 1A). Reference annotation combined the available fluorescence markers with tissue context: F4/80 identified Kupffer cells, Desmin identified stellate cells, and Phalloidin delineated cell borders and sinusoidal structures. Hepatocyte nuclei were identified from their characteristic morphology and position within Phalloidin-defined cell boundaries, whereas endothelial nuclei were assigned based on their elongated morphology, sinusoidal localization, and absence of F4/80 and Desmin signal (Fig. 1A). Nuclei not meeting the criteria for these four classes were assigned to the heterogeneous other-cell category. For classification, each segmented nucleus was extracted together with its instance mask and standardized to a common volumetric input, with only DAPI-derived information retained as model input (Fig. 1A; Supplementary Fig. S1A; Supp. Movie 1).

After quality-control filtering, the final dataset contained 14,385 nuclei (Table 1; Supplementary Fig. S1B). The class distribution was markedly imbalanced: hepatocytes comprised 7,089 nuclei (49.3%) and endothelial cells 4,923 (34.2%), whereas stellate cells (1,020; 7.1%), Kupffer cells (916; 6.4%), and other cells (437; 3.0%) were substantially less frequent. Thus, the most abundant class contained approximately 16-fold more nuclei than the least abundant class. Nuclear abundance and size distributions also varied across animals (Supplementary Fig. S1B,C), reinforcing the need to evaluate generalization at the biological-sample level rather than by randomly partitioning individual nuclei.

We therefore used five-fold leave-one-animal-out cross-validation, with one animal reserved as an independent test set in each fold and the remaining four animals forming the development set (Fig. 1C). Original nuclei were partitioned before augmentation. Within each development set, 10% of nuclei were reserved for validation using joint stratification by animal and class, while the remaining 90% were used for training. These fold-specific identities were shared across all modelling approaches. Validation data were used for classifier, hyperparameter, augmentation-level, and checkpoint selection, after which the selected model configurations were fixed and evaluated once on the held-out animal (Fig. 1C–E). Test nuclei were therefore excluded from training, augmentation, validation, and model selection.

For the feature-based approaches, training-only augmentation was evaluated using an original-only condition and nested class-balanced sampling targets of 100, 200, 500, 1000, 2000, and 4000 samples per class per training animal (Fig. 1D). Augmentation included geometric transformations, including axis permutations and flips, together with intensity scaling and shifting, Gaussian blurring, and additive Gaussian noise. Validation and test nuclei remained original and unaugmented. The direct 3D CNN baseline (ResNet3D-18) instead applied augmentation on the fly to original training nuclei during optimization. This design ensured that all reported predictive performance was measured exclusively on unaugmented nuclei from animals that were not used to fit or select the corresponding model.

Frozen 3DINO representations capture structured variation in hepatic nuclear morphotypes

From each standardized DAPI-only nuclear patch, we derived two complementary representations: a 31-dimensional handcrafted feature vector describing nuclear morphology, intensity, texture, and structural complexity, and a 1,024-dimensional embedding extracted with the frozen pretrained 3DINO model (Fig. 1D). We next examined the organization of these representations using principal component analysis (PCA) and uniform manifold approximation and projection (UMAP), restricting these analyses to original, unaugmented nuclei (Fig. 2). The 31-dimensional handcrafted representation was comparatively concentrated along its leading principal axes, with PC1 and PC2 explaining 53.16% and 11.97% of the variance, respectively (65.13% combined; Fig. 2A). By contrast, variance in the 1,024-dimensional frozen 3DINO representation was distributed across a larger number of components, with PC1 and PC2 accounting for 18.30% and 8.24% of the variance, respectively (26.54% combined; Fig. 2C; Supplementary Fig. S2A).

Both representations exhibited class-associated structure while retaining substantial overlap among nuclear morphotypes (Fig. 2). Hepatocyte nuclei occupied comparatively distinct regions in both representation spaces, whereas sinusoidal endothelial, stellate, Kupffer, and other-cell nuclei showed greater overlap. The nonlinear UMAP projections [19] further revealed local clustering within both representations without complete separation of the five reference morphotypes (Fig. 2B,D).

When the same representations were visualized according to animal of origin rather than morphotype, animal-associated structure was also apparent (Supplementary Fig. S2B,C). This observation further supports the use of animal-level data partitioning, as random nucleus-level splitting could allow sample-specific representation structure to be shared between training and test sets.

These dimensionality-reduction analyses were used exclusively to characterize representation structure and were not used for model selection or performance assessment. Quantitative conclusions regarding predictive generalization were therefore based on the held-out animal in each cross-validation fold.

Validation-based model selection yields fold-specific configurations

We next optimized the predictive models independently within each leave-one-animal-out fold. For both the frozen 3DINO embeddings and handcrafted nuclear features, we evaluated logistic regression, random forest, support vector machine (SVM), and multilayer perceptron (MLP) classifiers across the original-only condition and the six predefined augmentation levels. Model selection was performed exclusively on the validation subset, using macro-F1 as the primary criterion, with balanced accuracy and log loss used as predefined tie-breakers. The held-out animal was not used to select the classifier, augmentation level, hyperparameters, or training checkpoint.

Importantly, validation-based selection did not identify a single optimal configuration across all animals. For the 3DINO representation, SVM classifiers trained with 4000 samples per class per training animal were selected in two folds, whereas MLP classifiers were selected in the remaining three folds, using either 2000 or 4000 samples per class per training animal (Supplementary Fig. S3B). For the handcrafted representation, an MLP was selected in four folds, using either original-only training or augmentation targets of 1000 or 4000 samples per class per training animal, while a random forest trained with 2000 samples per class per training animal was selected in the remaining fold (Supplementary Fig. S3B). The direct 3D CNN baseline was likewise optimized independently within each fold using the same validation identities, with the selected ResNet3D-18 checkpoint occurring between epochs 10 and 26 (Supplementary Fig. S3A,B).

Across the predefined augmentation conditions, held-out performance showed an initial improvement after augmentation was introduced but did not increase monotonically with larger sampling targets (Supplementary Fig. S4). Instead, performance generally plateaued across the higher augmentation levels, with the preferred level varying among folds and classifier families. Importantly, these held-out augmentation curves were used only to characterize sensitivity to training-set augmentation and were not used to determine the selected procedures. Accordingly, all subsequent primary comparisons were based on the fold-specific validation-

selected procedure for each modelling approach rather than on a single classifier or augmentation level chosen from test performance.

Frozen 3DINO representations achieve the highest mean held-out performance

We next compared the three validation-selected modelling approaches on the five held-out animals: frozen 3DINO embeddings followed by a validation-selected classifier, handcrafted nuclear features followed by a validation-selected classifier, and direct supervised classification of the standardized DAPI patches using ResNet3D-18 (Fig. 3). Macro-F1 was used as the primary performance metric because it weights all five morphotypes equally, while balanced accuracy and weighted F1 were evaluated as complementary measures. Performance was summarized across animals with each held-out animal contributing equally to the reported mean and standard deviation.

The 3DINO-based procedure achieved the highest mean performance across all three metrics, with a macro-F1 of 0.596 ± 0.041, balanced accuracy of 0.625 ± 0.037, and weighted F1 of 0.831 ± 0.046 (Table 2; Fig. 3). The direct ResNet3D-18 classifier showed closely similar performance, reaching a macro-F1 of 0.584 ± 0.051, balanced accuracy of 0.613 ± 0.048, and weighted F1 of 0.821 ± 0.052. The handcrafted-feature procedure showed lower mean performance, with a macro-F1 of 0.511 ± 0.040, balanced accuracy of 0.547 ± 0.065, and weighted F1 of 0.775 ± 0.048.

For the primary metric, the mean paired difference in macro-F1 between the 3DINO and handcrafted procedures was 0.085, with a paired standardized effect size of $d_z = 2.773$. The corresponding raw Wilcoxon signed-rank p-value was 0.063 and the Holm-adjusted p-value was 0.562 (Supplementary Table S1). In contrast, the mean macro-F1 difference between 3DINO and ResNet3D-18 was only 0.012 ($d_z = 0.487$; raw p = 0.438); Holm-adjusted p = 0.875). A similar pattern was observed for balanced accuracy and weighted F1, for which the descriptive differences between 3DINO and the handcrafted representation were larger than those between 3DINO and ResNet3D-18 (Fig. 3; Supplementary Table S1).

None of the pairwise comparisons reached statistical significance after correction for multiple comparisons (all Holm-adjusted $p \geq 0.562$). Given that the analysis comprised only five independent held-out animals, these tests have limited power and were interpreted together with the paired effect sizes and animal-level score distributions. The results therefore support the conclusion that frozen 3DINO representations provide similar observed held-out performance to direct supervised 3D CNN training in this dataset, while yielding consistently higher mean performance than the handcrafted representation across the three-summary metrics.

Class-specific performance reveals persistent ambiguity among non-parenchymal morphotypes

Class-specific analysis showed pronounced differences in predictive performance among the five hepatic nuclear morphotypes (Fig. 4; Supplementary Table S2). Hepatocytes were classified most reliably by all three approaches, with pooled F1 scores of 0.995 for both 3DINO and ResNet3D-18 and 0.987 for the handcrafted-feature procedure. The principal classification challenge therefore lay in distinguishing the non-parenchymal and heterogeneous nuclear morphotypes rather than in separating hepatocytes from the remaining classes.

Among the non-parenchymal classes, the 3DINO procedure achieved pooled F1 scores of 0.454 for stellate cells, 0.611 for Kupffer cells, and 0.783 for endothelial cells. The corresponding ResNet3D-18 values were 0.436, 0.599, and 0.758, whereas the handcrafted-feature procedure achieved 0.254, 0.528, and 0.707, respectively (Supplementary Table S2). The largest descriptive difference between 3DINO and the handcrafted representation was therefore observed for stellate cells, while 3DINO and ResNet3D-18 showed broadly similar class-specific performance.

The pooled row-normalized confusion matrices further revealed that most classification errors occurred among the non-parenchymal morphotypes (Fig. 4). For the 3DINO procedure, 52.0% of stellate nuclei, 70.0% of Kupffer nuclei, and 72.4% of endothelial nuclei were assigned to their corresponding reference class.

Misclassified nuclei from these classes were frequently assigned to other non-parenchymal categories, indicating substantial overlap in the DAPI-derived nuclear features available to the classifiers.

The heterogeneous other-cell category remained particularly difficult for all three approaches, with pooled F1 scores of 0.159 for 3DINO, 0.161 for ResNet3D-18, and 0.160 for the handcrafted-feature procedure (Supplementary Table S2). Although ResNet3D-18 showed higher recall for this category than 3DINO (27.7% versus 19.5%), precision remained low, resulting in essentially similar F1 scores. For 3DINO, 43.9% of other-cell nuclei were assigned to the endothelial class (Fig. 4), consistent with the heterogeneous nature of this residual category and the limited discriminative information available from DAPI alone.

Because these pooled confusion matrices aggregate nuclei across animals, animals with larger numbers of nuclei contribute proportionally more observations. They were therefore treated as descriptive class-level summaries rather than as independent inferential comparisons. Animal-specific confusion matrices showed additional variation in the magnitude and pattern of errors across held-out animals (Supplementary Fig. S5), further supporting the use of animal-level performance as the primary basis for comparing modelling approaches.

3DINO representations partially overlap with handcrafted nuclear descriptors

To examine how the learned 3DINO representation relates to interpretable nuclear properties, we quantified associations between the 1,024-dimensional embeddings and the 31 handcrafted DAPI-derived descriptors using original, unaugmented nuclei (Fig. 5). Pearson correlation analysis showed that the leading principal components of the standardized 3DINO representation were associated with multiple descriptors of nuclear morphology, intensity, texture, and structural complexity (Fig. 5A). The strongest associations were concentrated among the leading components, whereas many later components showed weaker pairwise correlations with individual handcrafted variables.

Analysis of the individual 3DINO dimensions with the highest variance revealed a complementary pattern (Fig. 5B). Several embedding dimensions showed broad positive or negative associations with groups of handcrafted descriptors, whereas others exhibited more selective or weak relationships. Thus, the learned dimensions did not map in a simple one-to-one manner onto predefined nuclear measurements.

The overall pattern was also observed using rank-based Spearman correlations, and stratified analyses indicated that associations between the leading 3DINO components and handcrafted descriptors were not confined to a single animal or morphotype (Supplementary Fig. S6). Together, these results indicate that frozen 3DINO embeddings partially recapitulate variation captured by interpretable nuclear descriptors while also containing representation components that are only weakly associated with any individual handcrafted feature.

Because these analyses are correlational, they do not establish the biological meaning or independent predictive contribution of the weakly correlated embedding components. Their role is therefore best interpreted as evidence that the 3DINO representation is related to, but not reducible to, the predefined handcrafted feature space.

Discussion

This study establishes a marker-sparing framework for classifying hepatic 3D nuclear morphotypes from DAPI alone. Under five-fold leave-one-animal-out cross-validation, frozen 3DINO representations followed by fold-specific validation-selected classifiers achieved the highest mean macro-F1, balanced accuracy, and weighted F1 among the three evaluated approaches. However, the difference relative to direct ResNet3D-18 classification was small, whereas the separation from handcrafted DAPI-derived features was substantially larger. These results therefore do not support a claim of superiority over direct 3D CNN training, but show that a frozen pretrained volumetric representation can achieve a similar level of held-out animal performance without task-specific representation learning.

The central methodological finding is the successful cross-domain transfer of a self-supervised 3D representation pretrained outside fluorescence microscopy. 3DINO was developed from volumetric medical-imaging data

rather than microscopy of individual nuclei [15], yet its frozen embeddings retained sufficient information from standardized DAPI volumes to support hepatic nuclear morphotype classification. This extends the broader observation that self-supervised nuclear representations can capture phenotypically relevant information from DAPI images [14] by addressing a complementary question: whether representations learned from a substantially different volumetric imaging domain can transfer to nuclear phenotyping. The present results indicate that such transfer is feasible, at least within the imaging conditions and biological setting evaluated here.

The comparison with handcrafted nuclear descriptors further clarifies what is gained by the learned representation. Handcrafted features remain valuable because they provide explicit measurements of nuclear morphology, intensity, texture, and structural complexity and therefore offer a direct link between quantitative image analysis and interpretable nuclear properties. Nevertheless, the handcrafted-feature procedure showed lower mean performance than the 3DINO-based procedure across macro-F1, balanced accuracy, and weighted F1. The representation analysis provides a complementary perspective: several 3DINO components were associated with predefined nuclear descriptors, whereas others showed only weak or distributed pairwise relationships. Thus, the learned representation partially recapitulates established nuclear measurements but is not reducible to a simple combination of individual handcrafted features. Importantly, weak correlation with predefined descriptors does not by itself establish additional biological information or mechanistic relevance; the biological meaning of these representation components remains to be determined.

The comparison with ResNet3D-18 is particularly informative. Despite being pretrained outside the fluorescence-microscopy domain and remaining completely frozen during this study, 3DINO representations achieved similar observed held-out animal-level performance to a task-specific 3D CNN trained end-to-end on the hepatic nuclear dataset. The mean macro-F1 difference was only 0.012, and the paired comparison did not support a systematic performance difference between the two approaches. Thus, the value of the frozen representation is not that it clearly outperforms direct supervised CNN training, but that it reaches a comparable performance range while restricting task-specific optimization to a comparatively lightweight downstream classifier. This may be particularly useful in biological imaging settings with many segmented objects but relatively few independent biological samples.

The statistical comparisons require corresponding caution. The independent inferential unit was the held-out animal, yielding only five paired observations for each model comparison. Although the 3DINO procedure showed higher mean scores than the handcrafted procedure and large paired standardized effect sizes for several metrics, none of the pairwise comparisons remained significant after Holm correction. Differences between 3DINO and ResNet3D-18 were substantially smaller. The results should therefore be interpreted as descriptive evidence from this dataset rather than as statistical proof that one modelling strategy is generally superior. Larger cohorts will be required to estimate the magnitude and reproducibility of these differences with substantially greater precision.

Class-specific performance also highlights an important biological limitation of DAPI-only classification. Hepatocyte nuclei were identified with very high accuracy by all three approaches, whereas discrimination among stellate, Kupffer, endothelial, and other-cell nuclei was substantially more difficult. This pattern indicates that the principal challenge is not distinguishing the dominant hepatocyte population, but resolving morphotypes whose DAPI-derived nuclear characteristics overlap more strongly. The particularly low performance for the other-cell category is unsurprising given that it represents a residual group rather than a single coherent biological population. Accordingly, the predicted classes should be interpreted as marker-associated nuclear morphotypes rather than definitive molecular cell identities.

This distinction is central to the intended use of the approach. The framework is marker-sparing rather than marker-free: fluorescence markers and tissue-context information were required to establish the reference annotations used for model development and evaluation, whereas only DAPI-derived information was provided to the classifiers. The resulting predictions can therefore reduce dependence on dedicated cell-type channels when approximate morphotype annotation is sufficient, potentially preserving fluorescence channels for other

biological measurements. They should not, however, be interpreted as replacements for molecular markers when definitive cell identity is required.

Several limitations define the scope of these findings. First, only five animals were available, limiting statistical power despite the large number of nuclei. We addressed this by separating animals across cross-validation folds and treating the animal, rather than the nucleus, as the independent evaluation unit. Second, the data were generated using a single tissue-preparation and imaging workflow. Changes in staining intensity, voxel size, optical properties, microscope configuration, tissue processing, disease state, or laboratory environment could therefore introduce domain shifts that are not represented here. Third, the reference annotations themselves integrate positive-marker information with morphological and spatial criteria and should not be regarded as error-free molecular ground truth. Fourth, the heterogeneous other-cell category limits both predictive performance and biological interpretation. Finally, the representation analysis is correlational and cannot establish which learned features causally contribute to classification or what biological processes they encode.

Taken together, these results demonstrate that a frozen self-supervised volumetric representation pretrained outside fluorescence microscopy can transfer to DAPI-only hepatic nuclear morphotyping under strict animal-level evaluation. Importantly, the contribution is not that frozen 3DINO representations replace task-specific supervised learning, but that they provide a complementary strategy capable of reaching a similar observed performance range with substantially lighter task-specific training. Future studies should determine whether this advantage persists across larger cohorts, imaging platforms, tissue conditions, and external laboratories, and whether microscopy-specific adaptation or fine-tuning can further improve performance.

Methods

Animals, tissue preparation, staining and imaging

Five adult C57BL/6J mice were obtained from the animal facility of the Centro Regional de Estudios Avanzados para la Vida (CREAV), Universidad de Concepción, and maintained under specific-pathogen-free conditions with food provided ad libitum. All experimental procedures were approved by the Ethics and Biosecurity Committee for Research and Development of Universidad de Concepción (CICUA 2023–2025) and were conducted in accordance with relevant institutional guidelines and regulations and the ARRIVE guidelines.

Livers were fixed by intracardiac perfusion with 4% paraformaldehyde in phosphate-buffered saline (PBS) containing 0.1% Tween-20 and post-fixed overnight in the same solution at room temperature. Tissue sections of 100 µm thickness were obtained using a vibratome. Sections were stained with DAPI to visualize nuclei, Phalloidin to delineate F-actin-rich cell boundaries and sinusoidal structures, anti-F4/80 to identify Kupffer cells, and anti-Desmin to identify stellate cells. Antibodies, fluorophores, suppliers, catalogue numbers, and working dilutions are provided in Table 3. Following staining, tissue sections were optically cleared using an adapted SeeDB protocol [20].

Three-dimensional images were acquired using an inverted multiphoton laser-scanning microscope (Zeiss LSM780) equipped with a 40×/1.2 NA multi-immersion objective. Images were acquired at an isotropic voxel size of 0.3 × 0.3 × 0.3 µm. One central vein–portal vein (CV–PV) region from a randomly selected liver lobe was imaged from each animal, yielding five independent 3D volumes. Each volume covered approximately 500 × 250 × 100 µm, thereby sampling cellular organization along the lobular axis. The marker and structural channels were used to establish the reference annotations, whereas all predictive analyses described below used only the DAPI channel.

Image preprocessing, nuclear segmentation and reference annotation

Raw microscopy stacks were preprocessed to correct acquisition artifacts and standardize image quality across samples. Rapid image acquisition produced a serrated-border artifact caused by alternating shifts between adjacent Y-columns. To correct this distortion, even and odd columns were separated and aligned using a TV-L1 optical-flow method [21] implemented in scikit-image, followed by symmetric warping and recombination of the corrected columns. Channel alignment was subsequently performed in Fiji using the Correct 3D Drift procedure [22], with a fifth channel containing combined cell-border and nuclear signals used as the registration reference. To cover the complete central vein–portal vein (CV–PV) region, two partially overlapping image tiles were acquired and stitched using the Grid/Collection Stitching [23] plugin in Fiji [24] with 10% overlap. Border regions containing zero-valued pixels were subsequently cropped. Depth-dependent fluorescence attenuation along the z-axis was corrected in Fiji using cumulative histogram normalization, mapping intensities to the 10–99.99% range. Photobleaching was additionally corrected by histogram matching using the Fiji Bleach Correction plugin [25].

Following preprocessing, nuclei were segmented from the DAPI channel using NucVerse3D [16], generating three-dimensional instance-label volumes. Reference annotations were established manually using the fluorescence-marker channels together with nuclear morphology and spatial context. Hepatocyte nuclei were identified by their characteristic morphology and central position within Phalloidin-defined cell boundaries. Kupffer cells were identified by F4/80 labeling, whereas stellate cells were identified by Desmin labeling. Endothelial nuclei were assigned based on their elongated morphology and localization along Phalloidin-delineated sinusoidal structures, together with the absence of F4/80 and Desmin labeling. Nuclei that did not satisfy the criteria for these four morphotypes were assigned to the heterogeneous other-cell category, which included, among others, nuclei consistent with cholangiocytes, lymphocytes, and monocytes.

The resulting reference class maps were paired with the NucVerse3D instance-label volumes. For each segmented nucleus, the reference morphotype was assigned from the annotated class labels overlapping the corresponding nuclear instance, using the majority non-zero class within the mask. These reference annotations were used only to establish target labels for model development and evaluation; all predictive inputs used in the subsequent classification analyses were derived exclusively from the DAPI channel.

Nuclear patch extraction and standardization

For each segmented nucleus, the corresponding DAPI signal and instance mask were extracted from the preprocessed 3D volume. Nuclei containing fewer than 2500 voxels and nuclei located within 3 voxels of an image boundary were excluded to remove small or incompletely captured objects. DAPI volumes were normalized independently using the 30th and 99.999th intensity percentiles, with values clipped to the range [0,1]. Reference labels were assigned from the corresponding class map using the majority non-zero class within each nuclear instance mask; ambiguous assignments and excluded objects were recorded during quality control.

Each retained single-nucleus patch was subsequently min–max normalized and standardized to a common cubic representation. Patches smaller than 70 × 70 × 70 voxels were center-padded to this size; nuclei requiring a larger field of view were instead padded to the smallest cubic volume that fully enclosed the extracted nucleus. The resulting image and mask volumes were then resized to 112 × 112 × 112 voxels. DAPI images were resized using zero-order interpolation with antialiasing, whereas instance masks were resized using nearest-neighbor interpolation without antialiasing. The transformed instance mask was retained throughout downstream feature extraction to define the nuclear region.

The standardized patches provided a common input for all three modelling approaches. Inputs to 3DINO and ResNet3D-18 were mapped from [0,1] to [-1,1], whereas handcrafted intensity descriptors were calculated from the [0,1]-scaled patches. Only DAPI-derived image information and the corresponding nuclear instance mask

were used at this stage; fluorescence-marker channels used for reference annotation were excluded from all predictive inputs.

Cross-validation design and data partitioning

Predictive performance was evaluated using five-fold leave-one-animal-out cross-validation, reflecting the animal rather than the individual nucleus as the independent biological unit. In each fold, all nuclei from one animal were reserved as an independent test set, while nuclei from the remaining four animals constituted the development set. Within each development set, 10% of the original nuclei were reserved for validation, with the validation subset jointly stratified by animal and reference morphotype; the remaining nuclei were used for training. The validation count was rounded up to the next integer. Data partitioning was performed with a fixed random seed of 42.

Importantly, partitioning was performed at the level of original nuclei before any augmentation was generated. The same fold-specific training, validation, and test identities were used across the 3DINO, handcrafted-feature, and ResNet3D-18 approaches and across all augmentation conditions. Only nuclei assigned to the training subset were eligible for augmentation. Validation and test sets always contained original, unaugmented nuclei, preventing augmented versions of the same nucleus from appearing in more than one subset within a fold.

Validation data were used for model selection within each fold. Macro-F1 served as the primary selection criterion, with balanced accuracy and lower log loss used as predefined tie-breakers. For feature-based approaches, validation performance determined both classifier configuration and augmentation level; for ResNet3D-18, it determined hyperparameters and the selected training checkpoint. After selection, the fitted model was evaluated directly on the held-out animal without refitting on the combined training and validation subsets. The held-out test data were never used to select classifiers, augmentation levels, hyperparameters, or checkpoints. Selection was conducted independently in each fold, allowing the final configuration to differ between held-out animals.

Training-data augmentation

Training-data augmentation was applied only after the original nuclei had been assigned to the fold-specific training, validation, and test subsets. Thus, augmented variants could originate only from training nuclei, while validation and test nuclei remained original and unaugmented throughout model development and evaluation. For the 3DINO and handcrafted-feature approaches, we evaluated an original-only condition together with six predefined class-balanced augmentation targets of 100, 200, 500, 1000, 2000, and 4000 sampled patches per class per training animal. Sampling was performed with replacement from the available training nuclei. Consequently, at the largest augmentation level, each fold contained up to 80000 training samples (4 training animals × 5 classes × 4,000 samples), rather than 80000 distinct biological nuclei.

Augmentation included random axis permutations and independent flips along the spatial axes with probability p=0.3. Intensity augmentation comprised multiplicative scaling between 0.9 and 1.1 and additive shifting between −0.05 and 0.05, applied jointly with probability p=0.3. Gaussian blurring with =0.4–1.2 and additive Gaussian noise with =0.1 were each applied with probability p=0.3. Geometric transformations were applied identically to the DAPI image and its corresponding nuclear mask to preserve spatial correspondence.

The six augmentation levels were generated using nested sampling plans: the largest sampling plan was generated for each fold and representation family and progressively subsetted to obtain the smaller augmentation conditions. This ensured that comparisons across augmentation levels were based on nested rather than independently resampled training sets. Original, unaugmented representations were retained separately for validation, testing, the original-only baseline, and representation-space analyses.

For the direct ResNet3D-18 approach, augmentation was instead generated on the fly during network training from the original training nuclei, rather than by constructing the six finite augmented datasets. Validation and test images remained unaugmented, consistent with the feature-based approaches.

3DINO representation extraction

Self-supervised volumetric representations were extracted using the pretrained 3DINO model [15] as a frozen feature extractor. We used the high-resolution ViT3D configuration supplied with the 3DINO implementation and its pretrained weights. Standardized DAPI-only nuclear patches of 112x112x112 voxels were mapped from the [0,1] intensity range to [-1,1] before being passed to the network. No fluorescence-marker or structural channels were provided to the model.

The pretrained 3DINO backbone was kept frozen throughout the study and operated in evaluation mode; its parameters were not updated using the liver dataset. Each nuclear patch was therefore transformed directly into a fixed 1024-dimensional embedding, which served as the input to the downstream classical classifiers. Embeddings were extracted from both the original and augmented training patches, whereas validation and test embeddings were generated exclusively from the corresponding original, unaugmented nuclei. Feature extraction was performed on GPU in batches of 16, using eight data-loading workers.

This design isolates the transferability of the pretrained representation from task-specific representation learning: 3DINO itself was never fine-tuned on the hepatic microscopy data, and only the downstream classifier was trained for nuclear morphotype prediction.

Handcrafted nuclear feature extraction

As an interpretable DAPI-derived representation, 31 handcrafted nuclear features were extracted from the same standardized image patches and transformed instance masks used for 3DINO representation extraction. The feature set summarized complementary aspects of nuclear morphology, DAPI intensity distribution, three-dimensional texture, and finite-scale structural complexity. Texture descriptors were calculated from the masked three-dimensional DAPI volume rather than from two-dimensional projections, thereby retaining volumetric intensity organization. Among the morphological descriptors, mean nuclear radius was defined from the centroid-to-surface distance, whereas mean shape index was derived from local principal surface curvatures.

Intensity-based descriptors were calculated from DAPI values in the [0,1] range within the transformed nuclear mask. Morphological and geometric descriptors were calculated from the standardized 112x112x112-voxel representation and therefore describe geometry in standardized voxel coordinates, rather than the original physical dimensions of the nuclei. They should consequently not be converted to physical units by assigning the original 0.3-µm voxel size to the resized representation. Original nuclear volume in physical coordinates was retained separately during preprocessing for quality control.

The same feature-extraction procedure was applied to original and augmented training patches. Validation and test features were calculated exclusively from the corresponding original, unaugmented nuclei. No F4/80, Desmin, Phalloidin, or other non-DAPI fluorescence information contributed to the handcrafted feature vectors.

Feature-based classifiers and model selection

The same downstream classification framework was applied independently to the frozen 3DINO embeddings and the 31-dimensional handcrafted feature vectors. Four supervised classifier families were evaluated: logistic regression, random forest, support vector machine (SVM), and multilayer perceptron (MLP). For each representation family, classifiers were evaluated independently within each leave-one-animal-out fold across the original-only training condition and the six predefined augmentation levels.

Where applicable, feature standardization and principal component analysis (PCA) were incorporated into the classification pipeline. All preprocessing transformations, including scaling and predictive PCA, were fitted exclusively on the training data within the corresponding fold and subsequently applied unchanged to the validation and test sets. Classifier-specific hyperparameters, including regularization parameters, PCA settings,

SVM parameters, random-forest settings, and MLP architecture and optimization parameters, were evaluated using predefined search grids. MLP training used the shared fold-specific validation subset for early stopping.

For SVM models requiring class-probability estimates, probability calibration was performed using internal subsets of the training data. These internal calibration partitions were grouped by original nucleus identity, ensuring that augmented variants derived from the same nucleus could not appear simultaneously in the fitting and calibration subsets. The fold-level validation set used for model selection remained separate from this internal calibration procedure.

Model selection was performed independently for each representation family and cross-validation fold. Validation macro-F1 served as the primary selection criterion because it assigns equal weight to all five morphotype classes; balanced accuracy and lower log loss were used as predefined tie-breakers. Validation performance jointly determined the classifier configuration and, for the feature-based approaches, the augmentation level. The held-out animal was not inspected during this process. Once selection was complete, the already fitted selected model was evaluated once on the held-out animal without refitting on the combined training and validation data. Consequently, different classifier and augmentation configurations could be selected in different leave-one-animal-out folds.

Direct ResNet3D classification

As a direct supervised image-classification baseline, standardized DAPI-only nuclear patches were classified using a three-dimensional residual convolutional neural network based on the ResNet3D-18 architecture [26,27]. The network received the same 112x112x112-voxel DAPI patches used for 3DINO representation extraction, with intensities mapped to $[-1,1]$. The model comprised a 7x7x7 convolutional stem followed by four residual stages containing $[2,2,2,2]$ residual blocks, with an initial feature width of 32 channels. Adaptive average pooling, dropout, and a final fully connected layer produced logits for the five reference morphotypes.

The network was trained independently within each leave-one-animal-out fold using the same training, validation, and held-out test identities used for the feature-based approaches. Class imbalance was addressed using a class-weighted cross-entropy loss. Optimization was performed with AdamW, and the hyperparameter search evaluated dropout probabilities of 0.2 and 0.4, learning rates of 1×10^{-4} and 3×10^{-4}, and weight-decay values of 1×10^{-4} and 1×10^{-3}. The learning rate was adjusted during training using a *ReduceLROnPlateau* scheduler based on validation macro-F1. Training was performed for a maximum of 100 epochs, with early stopping after five consecutive epochs without improvement in validation performance. CUDA mixed-precision training was used, with a batch size of 256 and eight data-loading workers.

In contrast to the finite augmented datasets used for 3DINO and handcrafted-feature extraction, ResNet3D-18 received online augmentation generated from the original training nuclei during each training epoch. The augmentation operations followed the same general image transformations described above and were seeded reproducibly while varying across epochs. Validation and test nuclei were not augmented.

Model selection was performed exclusively from the fold-specific validation data. Validation macro-F1 was the primary selection criterion, with balanced accuracy and lower log loss used as predefined tie-breakers. The checkpoint with the selected validation performance was retained for each fold, and the corresponding held-out animal was evaluated only after model and checkpoint selection had been completed. No model was subsequently refitted using the combined training and validation sets. Thus, the direct CNN provided an independently optimized supervised baseline against which the frozen 3DINO representation approach could be compared under the same animal-level evaluation design.

Model evaluation and statistical analysis

Final predictive performance was assessed exclusively on the held-out animal in each of the five leave-one-animal-out folds. For each fold and methodological approach, evaluation loaded the model or checkpoint previously selected using validation data and generated predictions for all original, unaugmented nuclei from the corresponding test animal. The primary performance metric was macro-F1, selected because it weights each of the five morphotype classes equally despite the marked class imbalance. Balanced accuracy and weighted F1 were used as complementary summary measures, and overall accuracy was additionally reported. Class-specific precision, recall, and F1-score were calculated to characterize differences in performance among individual morphotypes.

The primary comparison involved the three independently validation-selected approaches: frozen 3DINO embeddings with a classical classifier, handcrafted nuclear features with a classical classifier, and direct ResNet3D-18 classification. Each animal contributed one held-out score per approach and therefore received equal weight in the animal-level summary statistics, irrespective of the number of nuclei contained in that animal. Accordingly, the independent inferential unit was the animal n=5, not the individual nucleus.

Pairwise differences between approaches were calculated within animals. For each performance metric, paired comparisons were assessed using the Wilcoxon signed-rank test, and paired standardized effect sizes d_z were calculated from the distribution of within-animal differences. Multiple-testing correction was performed using the Holm procedure across the predefined model pairs and the three principal comparison metrics: macro-F1, balanced accuracy, and weighted F1, within the corresponding comparison family. In addition, overall differences among the three approaches were summarized using model ranks and a within-animal permutation test based on the Friedman rank statistic.

Confusion matrices and one-vs-rest receiver-operating-characteristic curves were additionally generated from pooled held-out predictions to describe class-specific error patterns and discrimination. Unlike the animal-level summary metrics, these pooled analyses combine individual nuclei across test animals and therefore weight animals according to their numbers of nuclei. They were consequently treated as descriptive analyses, whereas formal comparisons among modelling approaches were based on the five paired animal-level observations.

Given the small number of independent animals and the overlapping development sets inherent to leave-one-animal-out cross-validation, inferential statistics were interpreted as exploratory. Statistical conclusions were therefore based jointly on paired effect sizes, the distribution of animal-level differences, and multiplicity-adjusted p values rather than on statistical significance alone.

Representation-space and correlation analyses

Representation-space analyses were performed using only original, unaugmented nuclei, with 3DINO embeddings and handcrafted feature vectors matched by nucleus identity. These analyses were conducted separately from predictive model fitting and were not used for classifier, hyperparameter, augmentation-level, or checkpoint selection.

The global organization of the two DAPI-derived representation spaces was examined using principal component analysis (PCA) and uniform manifold approximation and projection (UMAP) [19,28]. PCA was applied separately to the handcrafted-feature and 3DINO representation spaces after feature-wise standardization. UMAP was generated using 50 nearest neighbors, a minimum distance of 0.5, and random seed 42. Representations were visualized according to both reference morphotype and animal of origin to assess class-associated and animal-associated structure. PCA explained-variance ratios, component scores, and loadings were retained for quantitative inspection, whereas PCA and UMAP projections were treated exclusively as exploratory visualizations.

To examine the relationship between learned 3DINO representations and interpretable nuclear properties, the 1024-dimensional embedding space was compared with the 31 handcrafted DAPI-derived descriptors. For the primary analysis, Pearson correlation coefficients were calculated between the first 31 principal components of

the standardized 3DINO embeddings and each handcrafted feature. The proportion of total 3DINO variance explained by each principal component was retained to aid interpretation. A complementary analysis examined the 31 individual 3DINO embedding dimensions with the highest variance, with dimensions ranked according to their variance before standardization, and correlated these dimensions with the same handcrafted feature set.

As sensitivity analyses, Spearman rank correlations were calculated in addition to Pearson correlations, and correlation patterns were examined after stratification by animal and by reference morphotype. These analyses were used to determine whether observed global associations were broadly preserved across biological samples and classes rather than being attributable exclusively to a single animal or morphotype. Features with no variance within a given analysis subset were treated as undefined for correlation rather than being assigned an artificial value.

Data availability

The complete dataset and analysis outputs generated in this study are publicly available through Zenodo under https://doi.org/10.5281/zenodo.19502784. The deposited material includes the microscopy images, extracted single-nucleus image patches, trained models, and computational results used to generate the analyses reported in this study.

Code availability

The complete computational pipeline used for nuclear preprocessing and patch extraction, 3DINO representation extraction, handcrafted feature extraction, cross-validation assembly, feature-based model training and selection, ResNet3D-18 training, statistical analysis, representation-space analysis, and figure generation is publicly available at: https://github.com/BioDecodingLab/3D-HepaticNuclei-Classifier.

Acknowledgements

This work was supported by the Agencia Nacional de Investigación y Desarrollo (ANID), Chile, and Instituto Milenio Centro Interdisciplinario de Neurociencia de Valparaíso (CINV), under grant numbers ANID Fondecyt regular 1251048, and ICN2025_026 CINV to F.S-M. D.M-T. was supported by ANID-Subdirección de Capital Humano/Doctorado Nacional/2023-21230215.

Author contributions

A.T.B., F.S.-M. and H.M.-N. conceived the study. A.T.B., J.T., D.M-T. and F.S-M. contributed to sample preparation, imaging, annotation, and data generation. A.T.B. and H.M.-N. developed and evaluated the computational pipeline. A.T.B., F.S.-M. and H.M.-N. analyzed and interpreted the results. H.M.-N. drafted the manuscript with input from all authors. All authors reviewed, edited, and approved the final manuscript.

Ethics declarations

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process: The authors used ChatGPT during manuscript preparation for selected language revisions, stylistic refinement, code troubleshooting, and code refactoring. The tool was used as an assistive resource and did not replace scientific judgment or interpretation. All AI-assisted text and code changes were reviewed, checked, and revised by the authors, who remain fully responsible for the accuracy, reproducibility, and integrity of the work.

Competing interests

The authors declare no competing interests.

References

1. Nagy, P., Thorgeirsson, S. S. & Grisham, J. W. Organizational Principles of the Liver. in *The Liver* 1–13 (John Wiley & Sons, Ltd, 2020). doi:10.1002/9781119436812.ch1.
2. Fernández-Iglesias, A. *et al.* The hepatic sinusoid as a regulatory niche: From cellular specialization to precision hepatology. *Hepatology* https://doi.org/10.1097/HEP.0000000000001870 (2026) doi:10.1097/HEP.0000000000001870.
3. Martínez-Torres, D. *et al.* Phenotypic characterization of liver tissue heterogeneity through a next-generation 3D single-cell atlas. *Sci. Rep.* **14**, 2823 (2024).
4. Xu, J. *et al.* A spatiotemporal atlas of mouse liver homeostasis and regeneration. *Nat. Genet.* **56**, 953–969 (2024).
5. Morales-Navarrete, H. *et al.* A versatile pipeline for the multi-scale digital reconstruction and quantitative analysis of 3D tissue architecture. *eLife* **4**, e11214 (2015).
6. Hoehme, S., Friebel, A., Hammad, S., Drasdo, D. & Hengstler, J. G. Creation of Three-Dimensional Liver Tissue Models from Experimental Images for Systems Medicine. in *Hepatocyte Transplantation: Methods and Protocols* (eds Stock, P. & Christ, B.) 319–362 (Springer, New York, NY, 2017). doi:10.1007/978-1-4939-6506-9_22.
7. Hoehme, S. *et al.* Digital twin demonstrates significance of biomechanical growth control in liver regeneration after partial hepatectomy. *iScience* **26**, (2023).
8. Kingeter, L. M. & Schaefer, B. C. Expanding the multicolor capabilities of basic confocal microscopes by employing red and near-infrared quantum dot conjugates. *BMC Biotechnol.* **9**, 49 (2009).
9. Orth, A. *et al.* Super-multiplexed fluorescence microscopy via photostability contrast. *Biomed. Opt. Express* **9**, 2943–2954 (2018).
10. Damodaran, K., Crestani, M., Jokhun, D. S. & Shivashankar, G. V. Nuclear morphometrics and chromatin condensation patterns as disease biomarkers using a mobile microscope. *PLOS ONE* **14**, e0218757 (2019).

11. Kalinin, A. A. *et al.* 3D Shape Modeling for Cell Nuclear Morphological Analysis and Classification. *Sci. Rep.* **8**, 13658 (2018).
12. Rana, P., Sowmya, A., Meijering, E. & Song, Y. Estimation of three-dimensional chromatin morphology for nuclear classification and characterisation. *Sci. Rep.* **11**, 3364 (2021).
13. Morales-Navarrete, H., Nonaka, H., Segovia-Miranda, F., Zerial, M. & Kalaidzidis, Y. Automatic recognition and characterization of different non-parenchymal cells in liver tissue. in *2016 IEEE 13th International Symposium on Biomedical Imaging (ISBI)* 536–540 (2016). doi:10.1109/ISBI.2016.7493325.
14. Hua, Y., Li, S. & Zhang, Y. Self-supervised pretraining with NuSPIRe unlocks nuclear morphology-driven insights in spatial omics. *Genome Biol.* **27**, 71 (2026).
15. Xu, T. *et al.* A generalizable 3D framework and model for self-supervised learning in medical imaging. *Npj Digit. Med.* **8**, 639 (2025).
16. Vergara, J. *et al.* NucVerse3D: generalizable 3D nuclear instance segmentation across heterogeneous microscopy modalities. *Sci. Rep.* **16**, 22197 (2026).
17. Baratta, J. L. *et al.* Cellular organization of normal mouse liver: a histological, quantitative immunocytochemical, and fine structural analysis. *Histochem. Cell Biol.* **131**, 713–726 (2009).
18. Hildebrandt, F. *et al.* Spatial Transcriptomics to define transcriptional patterns of zonation and structural components in the mouse liver. *Nat. Commun.* **12**, 7046 (2021).
19. McInnes, L., Healy, J. & Melville, J. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. Preprint at https://doi.org/10.48550/arXiv.1802.03426 (2020).
20. Ke, M.-T., Fujimoto, S. & Imai, T. SeeDB: a simple and morphology-preserving optical clearing agent for neuronal circuit reconstruction. *Nat. Neurosci.* **16**, 1154–1161 (2013).
21. Zach, C., Pock, T. & Bischof, H. A Duality Based Approach for Realtime TV-L1 Optical Flow. in *Pattern Recognition* (eds Hamprecht, F. A., Schnörr, C. & Jähne, B.) 214–223 (Springer, Berlin, Heidelberg, 2007). doi:10.1007/978-3-540-74936-3_22.
22. Parslow, A., Cardona, A. & Bryson-Richardson, R. J. Sample Drift Correction Following 4D Confocal Time-lapse Imaging. *J. Vis. Exp. JoVE* e51086 (2014) doi:10.3791/51086.

23. Preibisch, S., Saalfeld, S. & Tomancak, P. Globally optimal stitching of tiled 3D microscopic image acquisitions. *Bioinformatics* **25**, 1463–1465 (2009).
24. Schindelin, J. *et al.* Fiji: an open-source platform for biological-image analysis. *Nat. Methods* **9**, 676–682 (2012).
25. Miura, K. Bleach correction ImageJ plugin for compensating the photobleaching of time-lapse sequences. *F1000Research* **9**, 1494 (2020).
26. Hara, K., Kataoka, H. & Satoh, Y. Learning Spatio-Temporal Features with 3D Residual Networks for Action Recognition. in *2017 IEEE International Conference on Computer Vision Workshops (ICCVW)* 3154–3160 (2017). doi:10.1109/ICCVW.2017.373.
27. He, K., Zhang, X., Ren, S. & Sun, J. Deep Residual Learning for Image Recognition. 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, 2016, pp. 770-778, doi: 10.1109/CVPR.2016.90.
28. Jolliffe, I. T. & Cadima, J. Principal component analysis: a review and recent developments. *Philos. Transact. A Math. Phys. Eng. Sci.* **374**, 20150202 (2016).

Tables

Animal	Hepatocyte	Stellate cell	Kupffer cell	Endothelial cell	Other cell	Total
Animal 1	1676	217	240	956	11	3100
Animal 2	1439	224	199	1081	96	3039
Animal 3	1615	169	220	1130	148	3282
Animal 4	1201	207	129	838	85	2460
Animal 5	1158	203	128	918	97	2504
All animals	7089	1020	916	4923	437	14385

Table 1. Composition of the final hepatic nuclear morphotype dataset by animal and reference class.

Numbers indicate nuclei retained after preprocessing and quality-control filtering for each of the five animals and five marker-assisted reference morphotypes. The final dataset comprised 14,385 nuclei: 7,089 hepatocytes, 1,020 stellate, 916 Kupffer, 4,923 endothelial, and 437 other-cell nuclei.

Representation / model family	Macro-F1	Balanced accuracy	Weighted F1	Accuracy	Animals
Frozen 3DINO + classifier	0.596±0.041	0.625±0.037	0.831±0.046	0.821±0.054	5
ResNet3D	0.584±0.051	0.613±0.048	0.821±0.052	0.803±0.061	5
Handcrafted + classifier	0.511±0.040	0.547±0.065	0.775±0.048	0.770±0.062	5

Table 2. Held-out animal-level classification performance of the three validation-selected modelling approaches.

Performance of frozen 3DINO representations followed by validation-selected classical classifiers, direct ResNet3D-18 classification, and handcrafted DAPI-derived nuclear features followed by validation-selected classical classifiers. Values are mean ± standard deviation across the five leave-one-animal-out test folds. Macro-F1 was the primary performance metric; balanced accuracy and weighted F1 were prespecified complementary metrics. Accuracy is additionally reported descriptively. Each animal contributes equally to the reported summary statistics.

Reagent type	Designation	Source	Identifier	Dilution	Description
Primary antibody	anti-F4/80 (Rat polyclonal)	Abcam	AB6640	1:400	Kupffer cell marker
Primary antibody	anti-Desmin (Rabbit monoclonal)	Abcam	ab15200	1:200	Stellate cell marker
Secondary antibody	Donkey anti-Rat CF 568	Biotium	20092	1:1000	—
Secondary antibody	Donkey anti-Rabbit Alexa Fluor 647	Invitrogen	A32795	1:1000	—
Small-molecule dye	Phalloidin Alexa Fluor 488	Invitrogen	A12379	1:1000	F-actin-rich cell boundaries
Small-molecule dye	DAPI	Invitrogen	D1306	1 µg/ml	Nuclei

Table 3. Antibodies and dyes used for tissue staining and reference annotation.

Reagents used to visualize nuclei, cell boundaries, and cell-type-associated markers in mouse liver tissue. F4/80 and Desmin were used for marker-assisted annotation of Kupffer and stellate cells, respectively. Phalloidin was used to delineate F-actin-rich cell boundaries, whereas DAPI provided the nuclear signal used as input for all classification approaches.

Figures

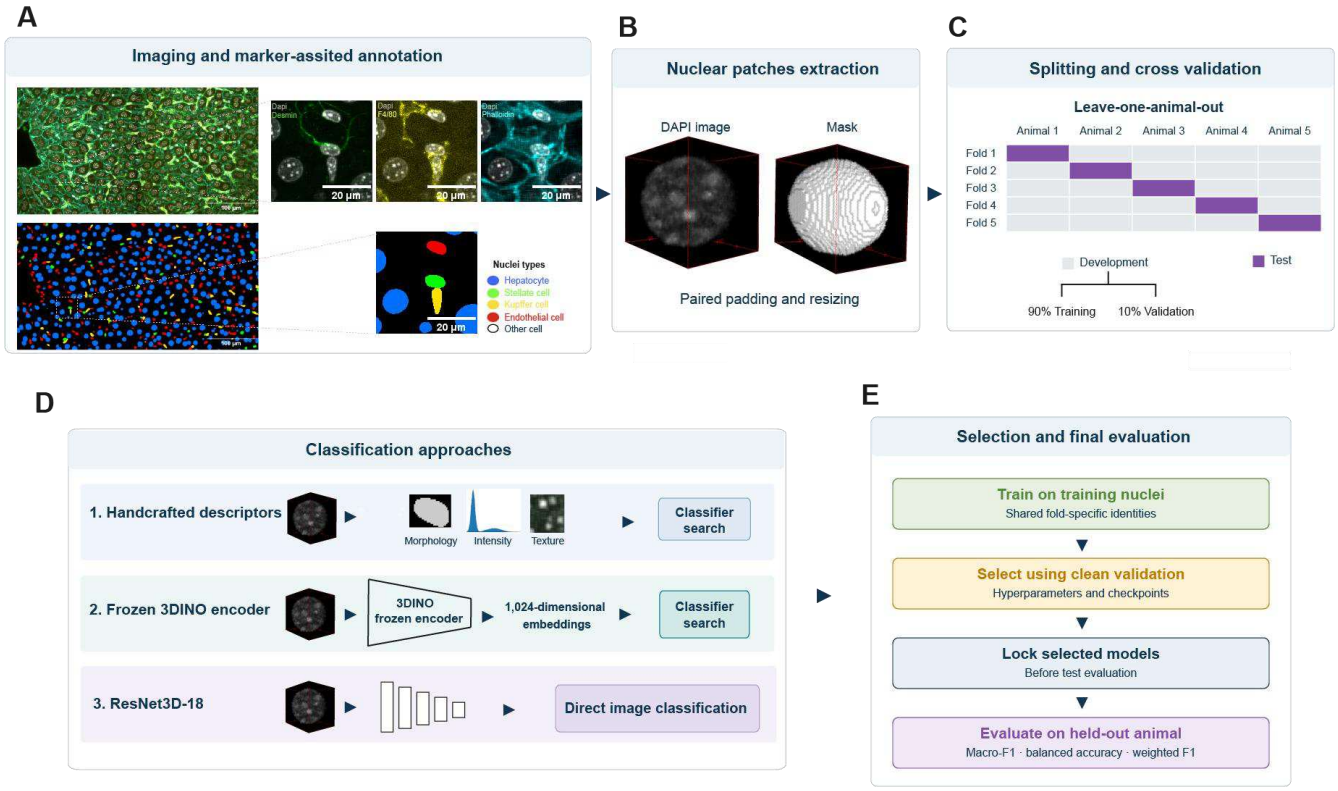


Figure 1. Experimental and computational workflow for DAPI-only classification of hepatic 3D nuclear morphotypes.
A) Representative three-dimensional mouse liver fluorescence microscopy data and marker-assisted reference annotation (maximum projection of 9 um). DAPI was used to visualize nuclei, F4/80 and Desmin supported identification of Kupffer and stellate cells, respectively, and Phalloidin provided cell-boundary and sinusoidal context. Segmented nuclei were assigned to five reference morphotypes: hepatocyte, stellate cell, Kupffer cell, endothelial cell, and other cell. **B)** Extraction and standardization of individual nuclear patches. For each segmented nucleus, the DAPI intensity patch and corresponding original instance mask were extracted as a paired input, padded when necessary, and resized to a common 112x112x112-voxel representation. **C)** Five-fold leave-one-animal-out cross-validation design. In each fold, one animal was reserved as the independent test set, while the remaining four animals formed the development set. Within the development set, 90% of the original nuclei were used for training and 10% for validation. Partitioning of original nuclei was performed before augmentation, and only training nuclei were eligible for augmentation. **D)** Three DAPI-derived classification strategies were compared: 31 handcrafted nuclear descriptors followed by classical classifier selection; frozen 3DINO encoding of DAPI patches into 1,024-dimensional embeddings followed by classical classifier selection; and direct end-to-end ResNet3D-18 classification of the standardized DAPI patches. **E)** Fold-specific model selection and final evaluation. Models were fitted using the training nuclei, while hyperparameters and checkpoints were selected exclusively using the validation subset. Selected models were then locked before test evaluation and applied to the held-out animal. Final performance was quantified using macro-F1, balanced accuracy, and weighted F1. Marker channels were used only to establish the reference annotations; all predictive inputs used DAPI-derived information only.

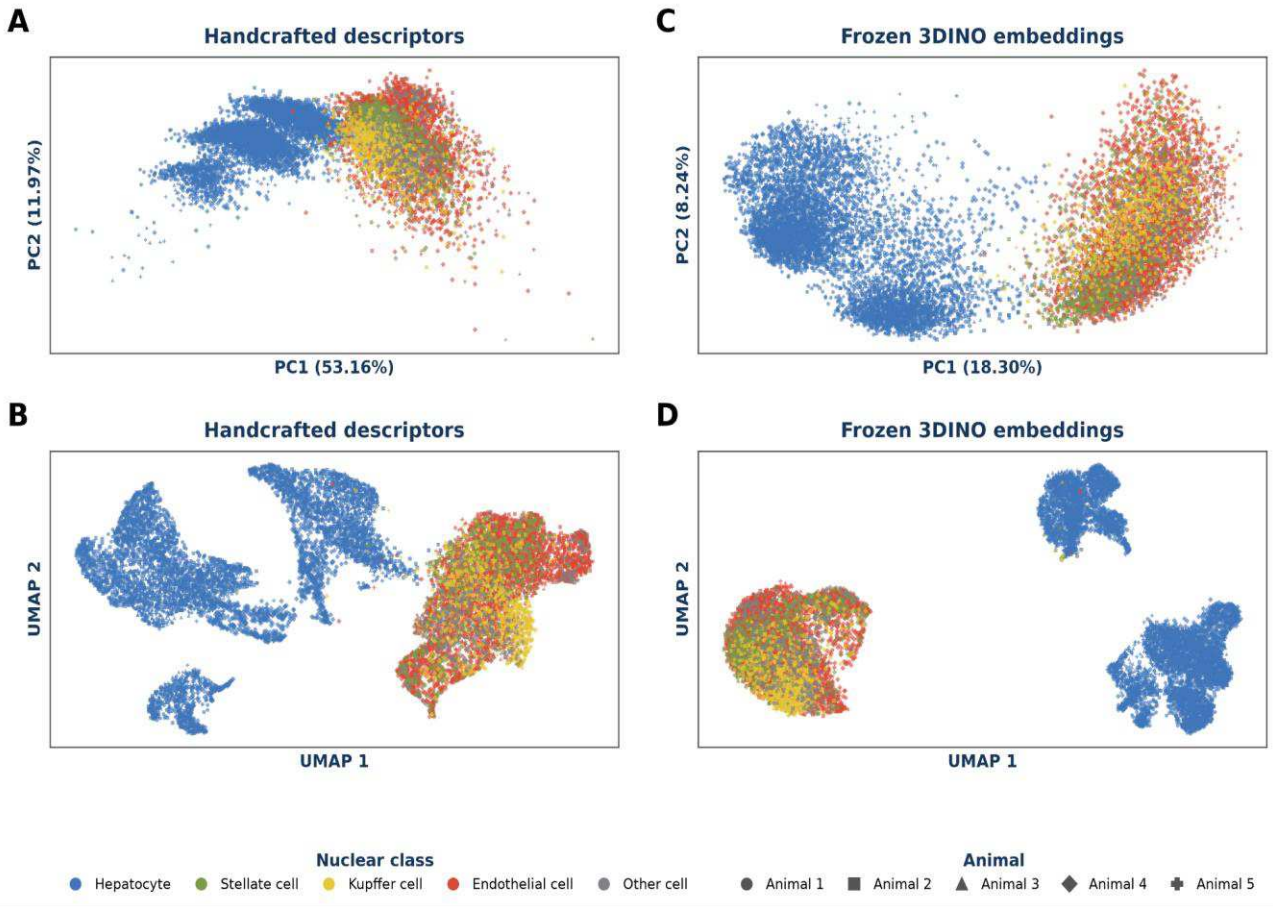


Figure 2. Representation structure of handcrafted nuclear features and frozen 3DINO embeddings. Principal component analysis (PCA) and uniform manifold approximation and projection (UMAP) were applied to original, unaugmented nuclei and were used only for exploratory representation analysis. **A)** PCA of the 31 handcrafted DAPI-derived nuclear features. PC1 and PC2 explained 53.16% and 11.97% of the total variance, respectively. **B)** UMAP projection of the handcrafted feature space. **C)** PCA of the 1,024-dimensional frozen 3DINO embeddings. PC1 and PC2 explained 18.30% and 8.24% of the total variance, respectively. **D)** UMAP projection of the 3DINO representation space. Each point represents one nucleus and colors denote marker-assisted reference morphotypes. The projections were not used for model selection or quantitative performance assessment.

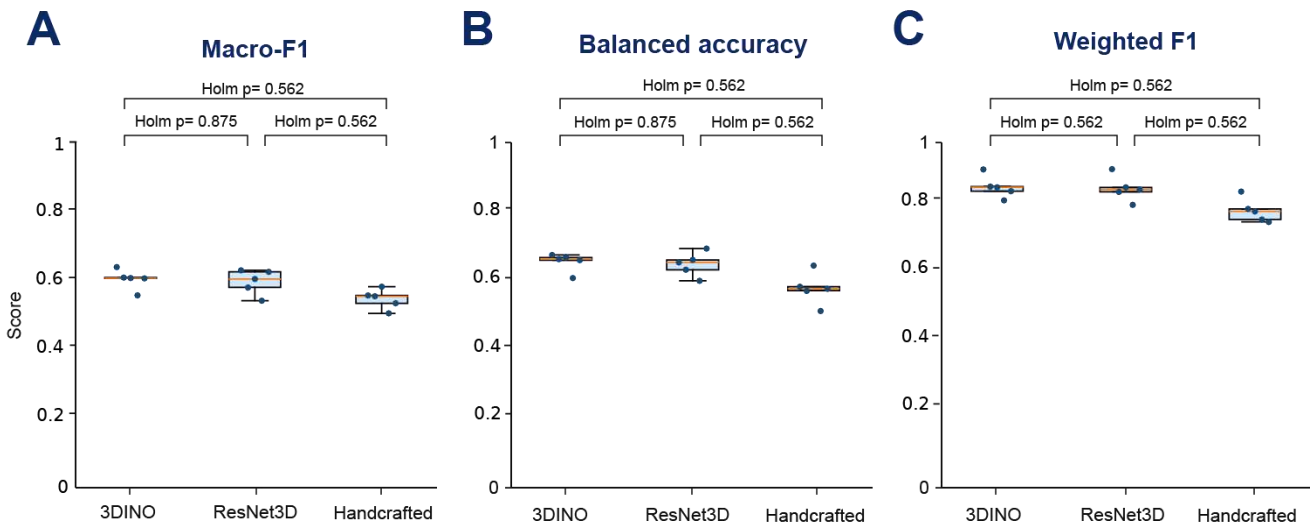


Figure 3. Held-out animal-level performance of the three validation-selected classification approaches. Performance was assessed independently on the five held-out animals using **A)** macro-F1, **B)** balanced accuracy, and **C)** weighted F1. The compared approaches were frozen 3DINO representations followed by fold-specific validation-selected classical classifiers, direct ResNet3D-18 classification, and handcrafted DAPI-derived nuclear features followed by fold-specific validation-selected classical classifiers. Each point represents one held-out animal ($n = 5$); animals therefore contribute equally irrespective of their numbers of nuclei. Macro-F1 was the primary metric. Pairwise brackets report Holm-adjusted two-sided Wilcoxon signed-rank test *P* values calculated from paired animal-level scores. None of the pairwise comparisons remained significant after multiple-testing correction.

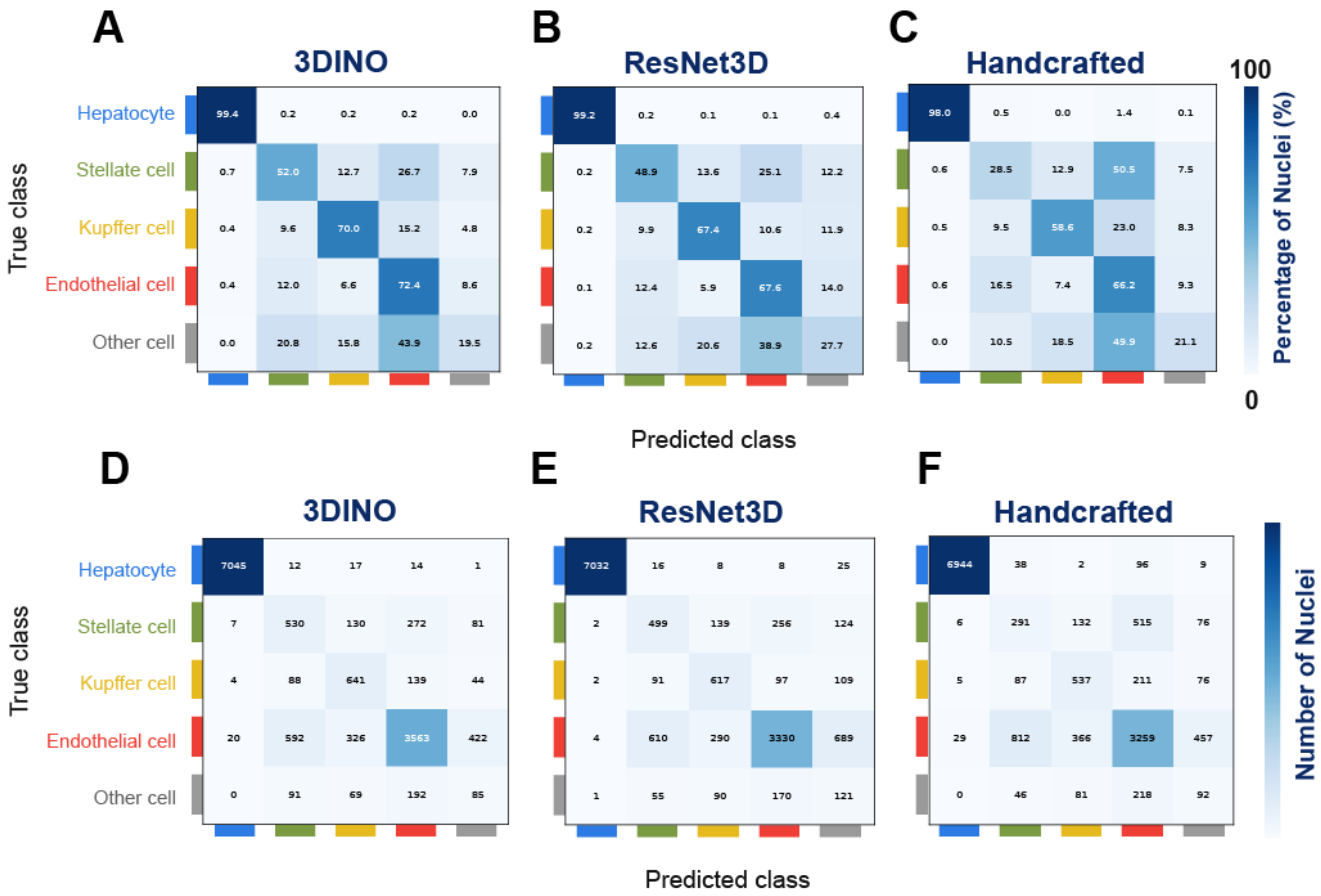
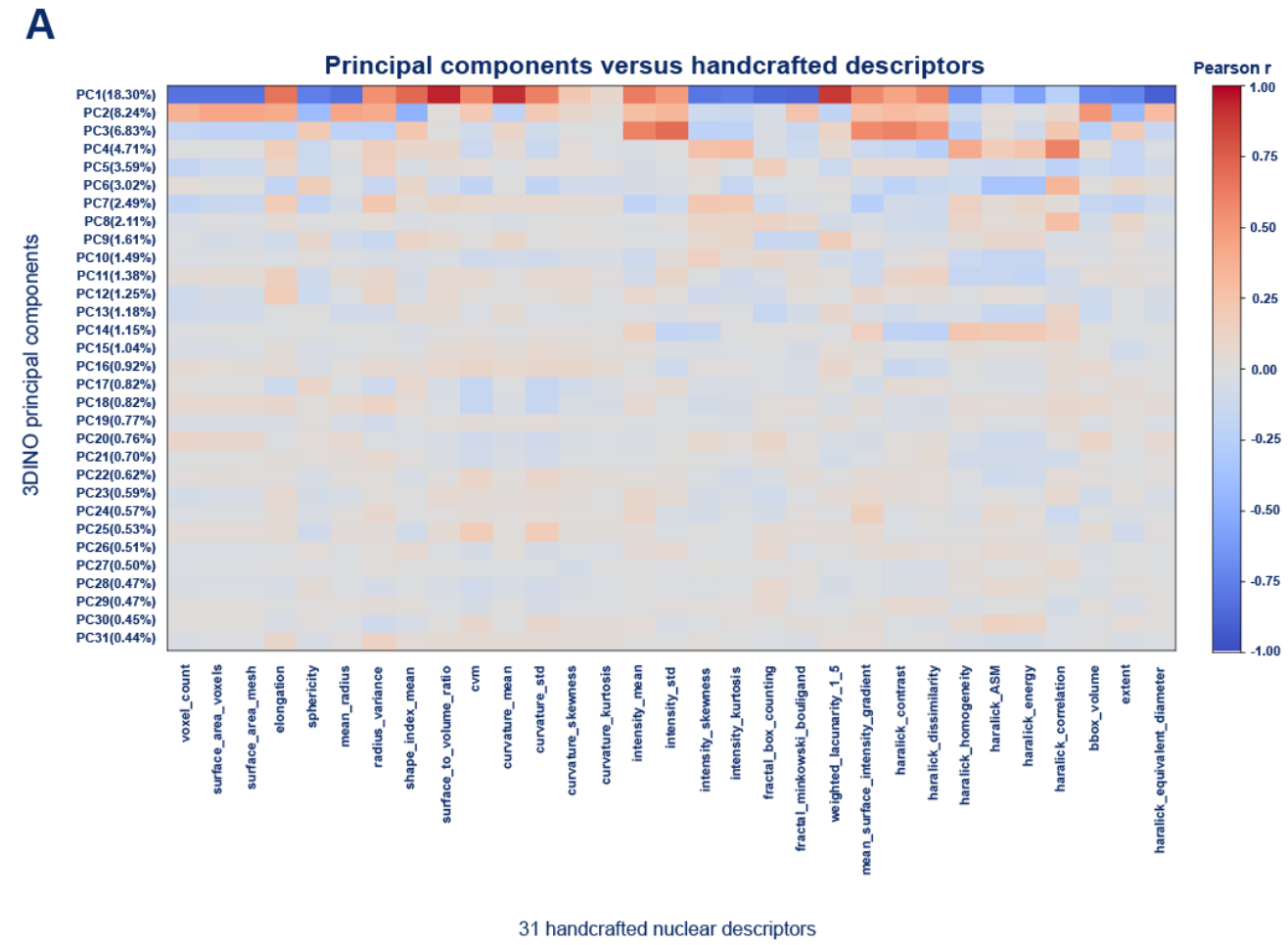


Figure 4. Class-specific prediction patterns of the validation-selected approaches. Row-normalized confusion matrices summarize pooled held-out predictions across the five leave-one-animal-out test folds for **A)** frozen 3DINO representations with fold-specific validation-selected classifiers, **B)** ResNet3D-18, and **C)** handcrafted nuclear features with fold-specific validation-selected classifiers. Rows indicate marker-assisted reference morphotypes and columns indicate predicted morphotypes. Values represent the proportion of nuclei from each reference class assigned to each predicted class. Because predictions are pooled across nuclei, these matrices weight animals according to their numbers of nuclei and are therefore presented as descriptive class-level analyses rather than animal-level inferential comparisons.


A
Principal components versus handcrafted descriptors
Pearson r
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
3DINO principal components
PC1(18.30%)
PC2(8.24%)
PC3(6.83%)
PC4(4.71%)
PC5(3.59%)
PC6(3.02%)
PC7(2.49%)
PC8(2.11%)
PC9(1.61%)
PC10(1.49%)
PC11(1.38%)
PC12(1.25%)
PC13(1.18%)
PC14(1.15%)
PC15(1.04%)
PC16(0.92%)
PC17(0.82%)
PC18(0.82%)
PC19(0.77%)
PC20(0.76%)
PC21(0.70%)
PC22(0.62%)
PC23(0.59%)
PC24(0.57%)
PC25(0.53%)
PC26(0.51%)
PC27(0.50%)
PC28(0.47%)
PC29(0.47%)
PC30(0.45%)
PC31(0.44%)
voxel_count
surface_area_voxels
surface_area_mesh
elongation
sphericity
mean_radius
radius_variance
shape_index_mean
surface_to_volume_ratio
cvm
curvature_mean
curvature_std
curvature_skewness
curvature_kurtosis
intensity_mean
intensity_std
intensity_skewness
intensity_kurtosis
fractal_box_counting
fractal_minkowski_bouligand
weighted_lacunarity_1_5
mean_surface_intensity_gradient
haralick_contrast
haralick_dissimilarity
haralick_homogeneity
haralick_ASM
haralick_energy
haralick_correlation
bbox_volume
extent
haralick_equivalent_diameter
31 handcrafted nuclear descriptors

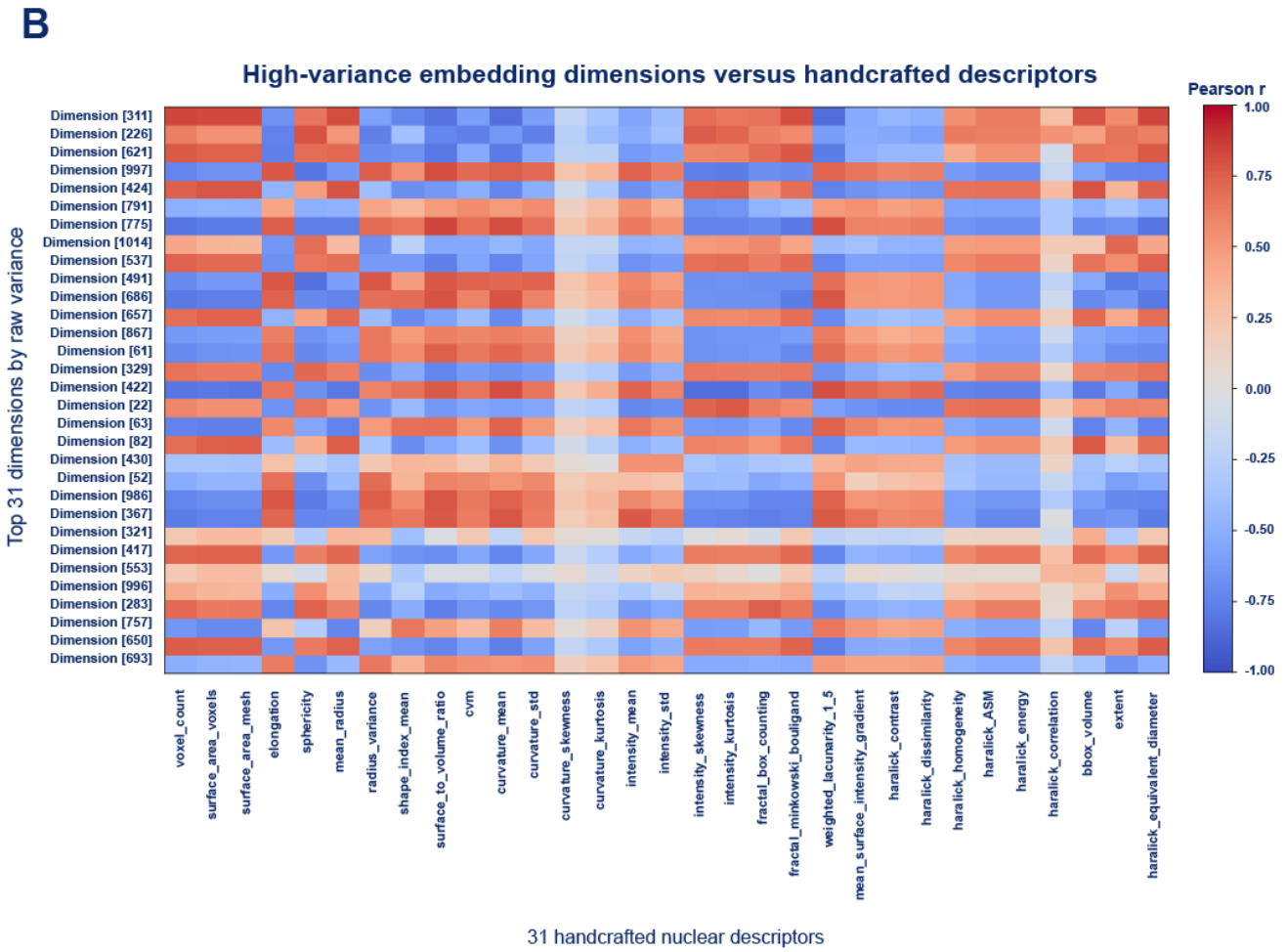

B
High-variance embedding dimensions versus handcrafted descriptors
Pearson r
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
Top 31 dimensions by raw variance
Dimension [311]
Dimension [226]
Dimension [621]
Dimension [997]
Dimension [424]
Dimension [791]
Dimension [775]
Dimension [1014]
Dimension [537]
Dimension [491]
Dimension [686]
Dimension [657]
Dimension [867]
Dimension [61]
Dimension [329]
Dimension [422]
Dimension [22]
Dimension [63]
Dimension [82]
Dimension [430]
Dimension [52]
Dimension [986]
Dimension [367]
Dimension [321]
Dimension [417]
Dimension [553]
Dimension [996]
Dimension [283]
Dimension [757]
Dimension [650]
Dimension [693]
voxel_count
surface_area_voxels
surface_area_mesh
elongation
sphericity
mean_radius
radius_variance
shape_index_mean
surface_to_volume_ratio
cvm
curvature_mean
curvature_std
curvature_skewness
curvature_kurtosis
intensity_mean
intensity_std
intensity_skewness
intensity_kurtosis
fractal_box_counting
fractal_minkowski_bouligand
weighted_lacunarity_1_5
mean_surface_intensity_gradient
haralick_contrast
haralick_dissimilarity
haralick_homogeneity
haralick_ASM
haralick_energy
haralick_correlation
bbox_volume
extent
haralick_equivalent_diameter
31 handcrafted nuclear descriptors

Figure 5. Relationship between frozen 3DINO representations and handcrafted nuclear descriptors. Pearson correlation analysis was performed using identity-matched original, unaugmented nuclei. **A)** Correlations between the first 31 principal components of the standardized 3DINO embedding space and the 31 handcrafted DAPI-derived nuclear descriptors. The percentage of 3DINO embedding variance explained by each principal component is indicated for the corresponding component. **B)**Correlations between the 31 individual 3DINO embedding dimensions with the highest variance and the same 31 handcrafted descriptors; embedding dimensions were ranked by variance before standardization. The color scale indicates the direction and magnitude of the Pearson correlation coefficient. These analyses assess correspondence between learned and predefined representations and were not used for predictive model selection.

Supplementary tables

Metric	Left family	Right family	Animals	Mean difference	Paired d_z	Raw p	Holm p
Macro-F1	3DINO	Handcrafted	5	0.085	2.773	0.063	0.562
	3DINO	ResNet3D	5	0.012	0.487	0.438	0.875
	Handcrafted	ResNet3D	5	-0.073	-1.701	0.063	0.563
Balanced accuracy	3DINO	Handcrafted	5	0.078	1.845	0.063	0.563
	3DINO	ResNet3D	5	0.012	0.358	0.438	0.875
	Handcrafted	ResNet3D	5	-0.066	-1.285	0.125	0.563
Weighted F1	3DINO	Handcrafted	5	0.056	3.174	0.063	0.563
	3DINO	ResNet3D	5	0.010	0.830	0.125	0.563
	Handcrafted	ResNet3D	5	-0.046	-1.630	0.063	0.563

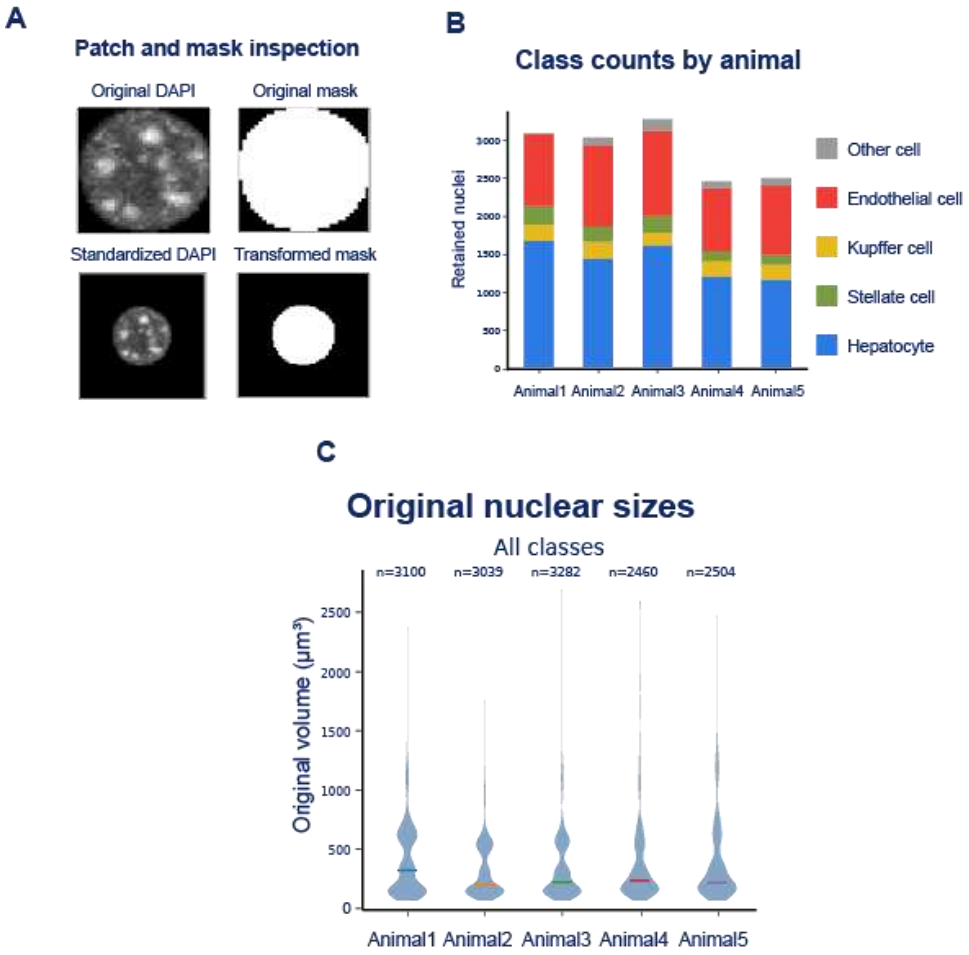
Supplementary Table S1. Paired animal-level statistical comparisons among the three validation-selected classification approaches.
Pairwise comparisons are reported for macro-F1, balanced accuracy, and weighted F1 across the five held-out animals. For each comparison, the table reports the mean paired difference, paired standardized effect size (d_z), raw two-sided Wilcoxon signed-rank test P value, and Holm-adjusted P value. Multiplicity correction was applied across the prespecified model-pair and metric comparison family. Positive differences indicate higher scores for the first-listed approach. The animal was the independent statistical unit ($n = 5$).

Family	Nuclear class	Precision	Recall	F1	Support
Handcrafted	Hepatocyte	0.994	0.980	0.987	7089.0
	Stellate cell	0.228	0.285	0.254	1020.0
	Kupffer cell	0.480	0.586	0.528	916.0
	Endothelial cell	0.758	0.662	0.707	4923.0
	Other cell	0.130	0.211	0.160	437.0
3DINO	Hepatocyte	0.996	0.994	0.995	7089.0
	Stellate cell	0.404	0.520	0.454	1020.0
	Kupffer cell	0.542	0.700	0.611	916.0
	Endothelial cell	0.852	0.724	0.783	4923.0
	Other cell	0.134	0.195	0.159	437.0
ResNet3D	Hepatocyte	0.999	0.992	0.995	7089.0
	Stellate cell	0.393	0.489	0.436	1020.0
	Kupffer cell	0.539	0.674	0.599	916.0
	Endothelial cell	0.862	0.676	0.758	4923.0
	Other cell	0.113	0.277	0.161	437.0

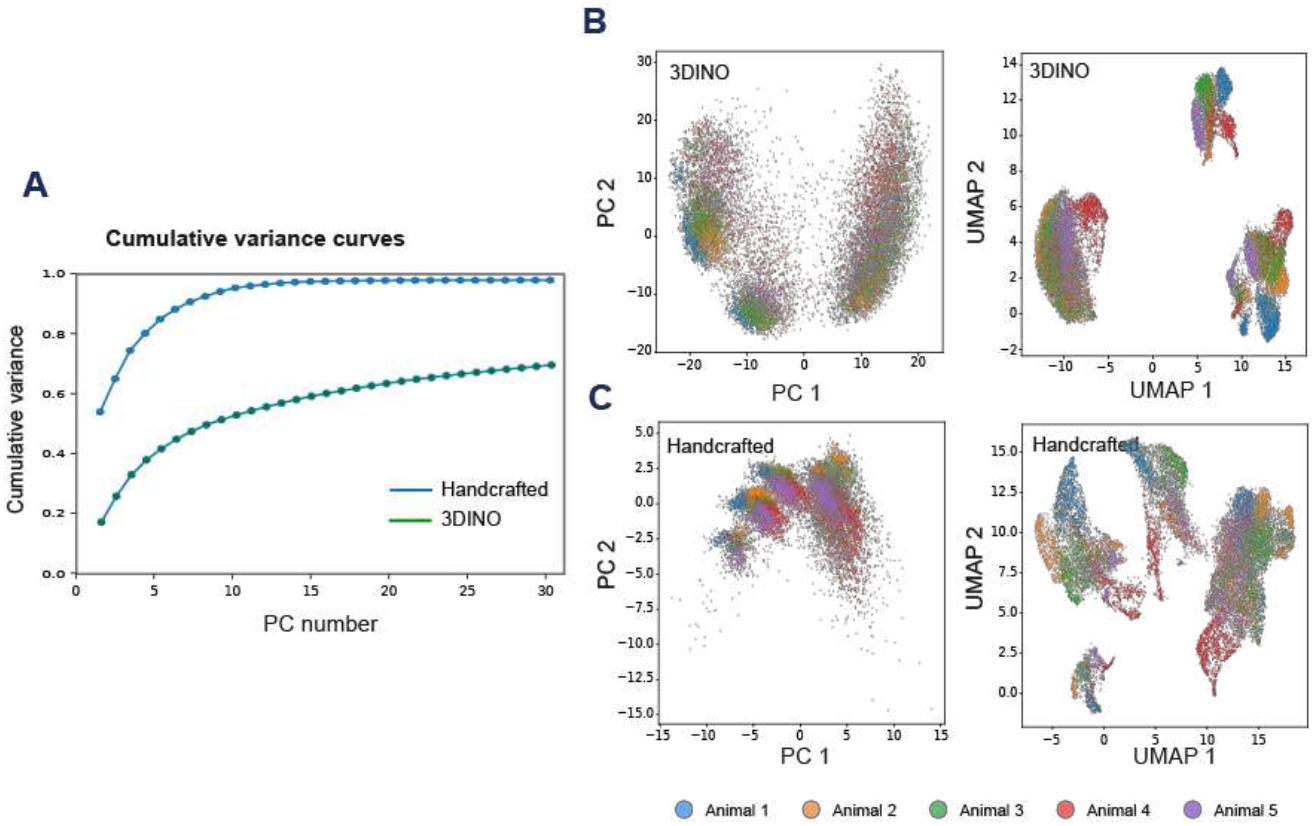
Supplementary Table S2. Pooled class-specific performance of the three validation-selected classification

approaches.
Precision, recall, F1-score, and support are reported for hepatocyte, stellate, Kupffer, endothelial, and other reference morphotypes using pooled held-out predictions from the five leave-one-animal-out folds. Results are shown for frozen 3DINO representations with fold-specific validation-selected classifiers, ResNet3D-18, and handcrafted nuclear features with fold-specific validation-selected classifiers. Because nuclei are pooled across animals, these values are descriptive and weigh animals according to their numbers of nuclei.

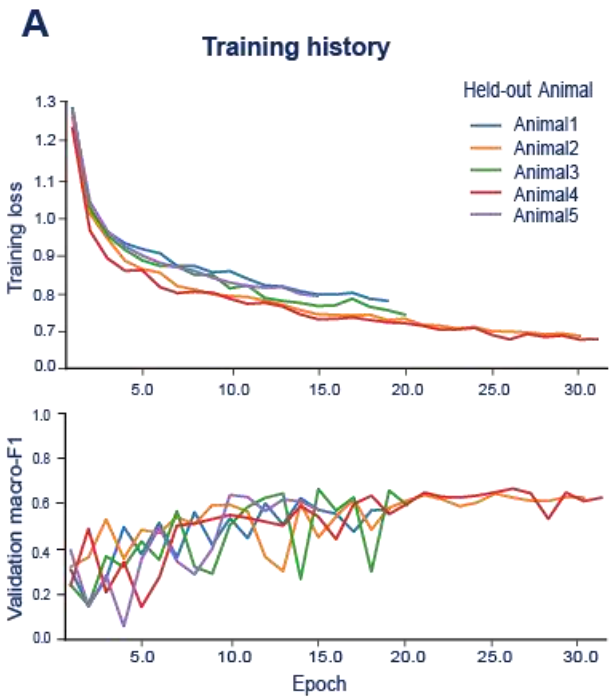
Supplementary figures



Supplementary Figure S1. Dataset composition and quality control of extracted nuclear patches. **A)** Representative examples of extracted and standardized DAPI-only nuclear patches and corresponding instance masks after spatial standardization. **B)** Numbers of retained nuclei across animals and marker-assisted reference morphotypes after preprocessing and quality-control filtering. **C)** Distribution of nuclear size measurements across the retained dataset, illustrating variation among animals and reference morphotypes. These quality-control analyses characterize the input data used for subsequent feature extraction and classification.



Supplementary Figure S2. Additional representation-space analyses of handcrafted and 3DINO features. Representation analyses were performed using original, unaugmented nuclei only. **A)** Explained-variance profile of principal components derived from the standardized 3DINO embedding space and handcrafted nuclear feature space, complementing the two-dimensional PCA projection in Fig. 2. **B)** Low-dimensional organization of the frozen 3DINO representation space visualized according to animal of origin using PCA and UMAP. **C)** Corresponding organization of handcrafted nuclear feature space visualized according to animal of origin. The animal-associated structure visible in these exploratory projections motivated evaluation at the animal rather than nucleus level. These analyses were not used for model fitting or selection.

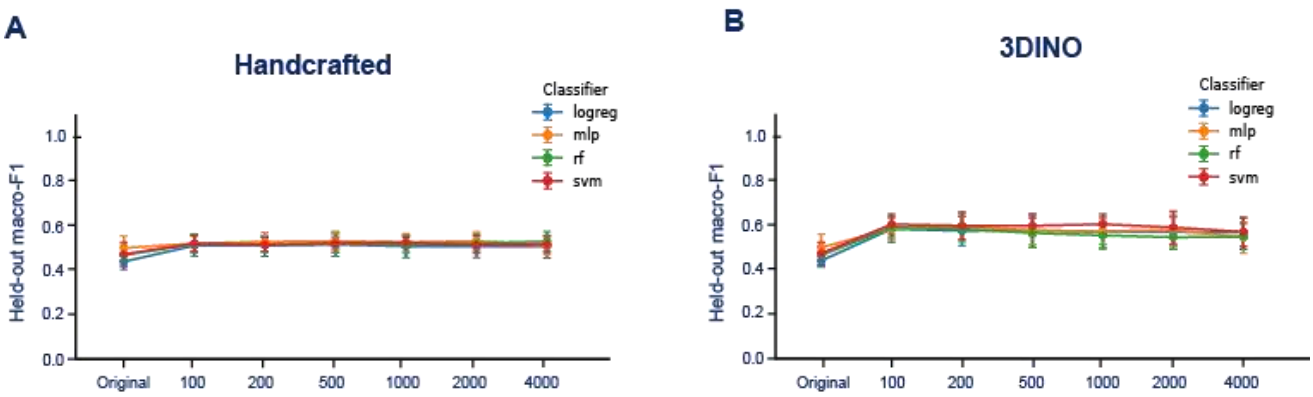


B

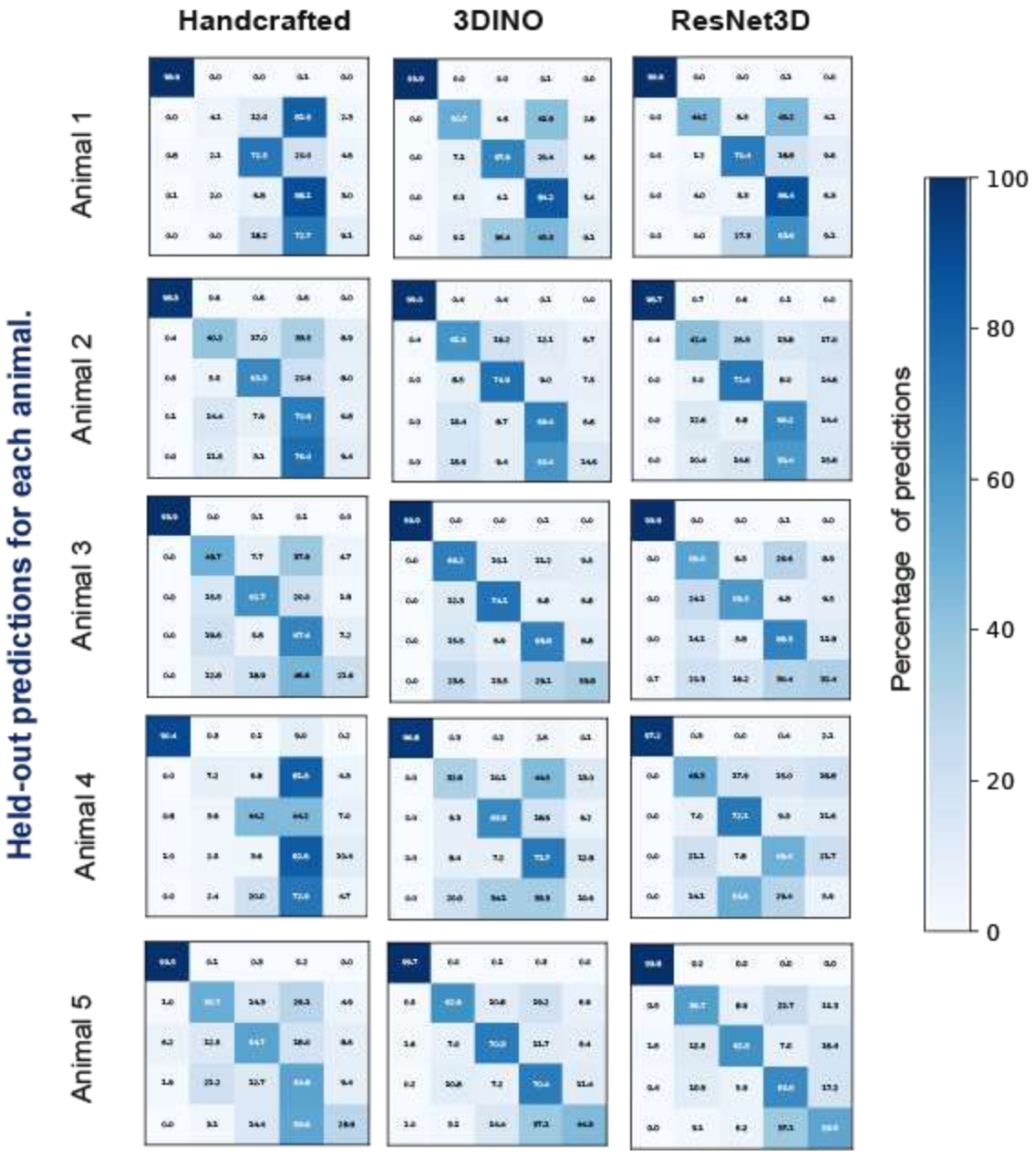
Selection by fold

Animal	Handcrafted	3DINO	ResNet3D
Animal1	MLP-N/A	SVM-4000	ResNet3D_18 14EPOCH
Animal2	MLP-4000	SVM-4000	ResNet3D_18 25EPOCH
Animal3	MLP-4000	MLP-2000	ResNet3D_18 15EPOCH
Animal4	RF-2000	MLP-4000	ResNet3D_18 26EPOCH
Animal5	MLP-1000	MLP-4000	ResNet3D_18 10EPOCH

Supplementary Figure S3. Fold-specific model selection and ResNet3D-18 training behavior. **A)** Validation performance trajectories for ResNet3D-18 training, with the selected checkpoint indicated for each fold. Checkpoints and hyperparameters were selected using validation data only; the held-out animal was not used during model selection. **B)** Fold-specific configurations selected from validation data for the frozen 3DINO and handcrafted-feature approaches, including classifier family and training-data augmentation level. Because selection was performed independently within each leave-one-animal-out fold, different configurations could be selected for different held-out animals.

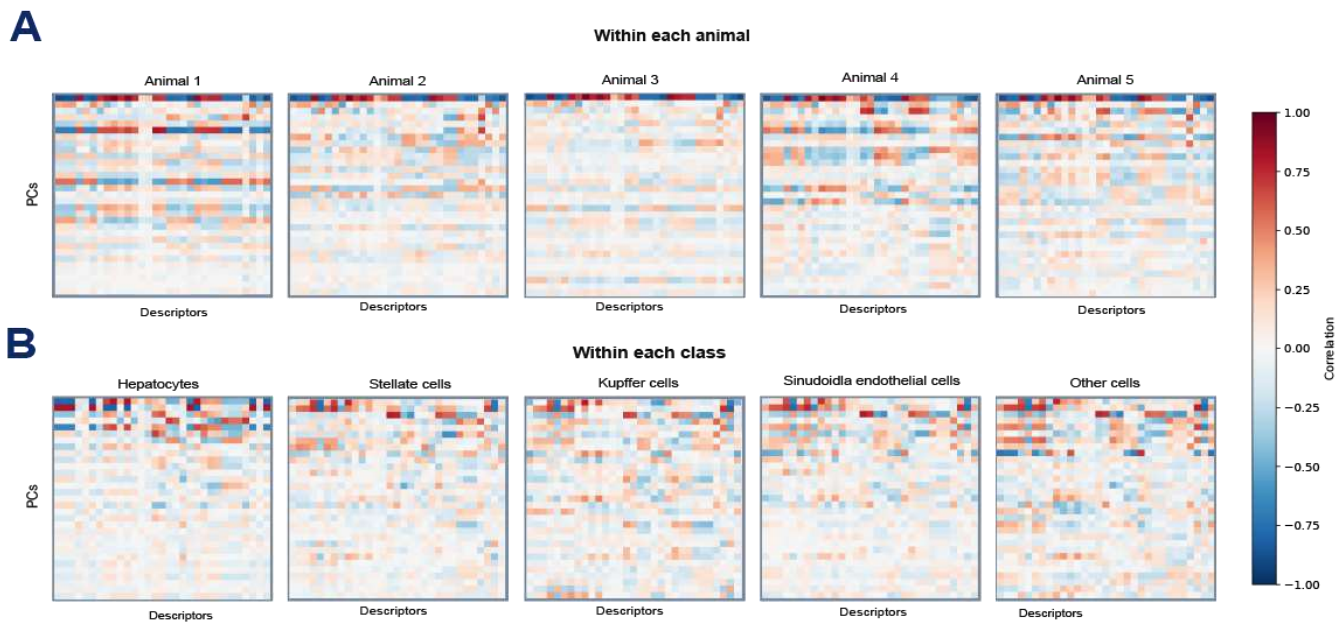


Supplementary Figure S4. Sensitivity of held-out macro-F1 to training-data augmentation level. Held-out macro-F1 was evaluated across the original-only training condition and six finite augmentation targets of 100, 200, 500, 1,000, 2,000, and 4,000 sampled training patches per class per training animal. **A)** Classifiers trained on the 31 handcrafted DAPI-derived nuclear descriptors. **B)** Classifiers trained on frozen 1,024-dimensional 3DINO embeddings. Curves correspond to logistic regression (LR), multilayer perceptron (MLP), random forest (RF), and support vector machine (SVM). Points indicate the mean macro-F1 across the five leave-one-animal-out held-out test folds, and error bars indicate the standard deviation across animals. Augmentation was applied only to training nuclei; validation and held-out test nuclei remained original and unaugmented. The held-out test curves are shown descriptively to illustrate sensitivity to training-set augmentation and were not used for model or augmentation-level selection, which was performed exclusively using the fold-specific validation sets.



Supplementary Figure S5. Animal-specific confusion matrices for the three validation-selected approaches.

Row-normalized confusion matrices show predictions separately for each held-out animal rather than pooling nuclei across folds. Columns correspond to the five held-out animals and rows correspond to the three modelling approaches: frozen 3DINO representations with fold-specific validation-selected classifiers, ResNet3D-18, and handcrafted nuclear features with fold-specific validation-selected classifiers. Within each matrix, rows indicate marker-assisted reference morphotypes and columns indicate predicted morphotypes. These matrices illustrate between-animal variability in class-specific prediction patterns.



Supplementary Figure S6. Rank-based and stratified correlations between 3DINO representations and handcrafted nuclear descriptors.
Sensitivity analyses complementing the Pearson correlations shown in Fig. 5. Spearman rank correlations between leading 3DINO principal components and the 31 handcrafted DAPI-derived nuclear descriptors: **A)** Correlation patterns stratified by animal of origin. **B)** Correlation patterns stratified by marker-assisted reference morphotype. Analyses used identity-matched original, unaugmented nuclei. Undefined correlations arising from constant features within a subset were retained as undefined rather than assigned an artificial numerical value.

Supplementary Movie 1. Three-dimensional visualization of marker-assisted hepatic nuclear morphotypes.

Rotating 3D renderings of a representative mouse liver volume showing the complete segmented nuclear population followed by the five marker-assisted reference morphotypes used in this study: hepatocyte, stellate, Kupffer, endothelial, and other cells. The sequence illustrates the spatial distribution and characteristic nuclear organization of each class within the imaged liver volume. Marker channels were used to establish the reference annotations, whereas subsequent classification analyses used only DAPI-derived nuclear information.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- Suppmovie1.mp4